

- 13.7 (a)** Show that the Reimers mass loss rate, given by Eq. (13.13) in Problem 13.6, can also be written in the form

$$\dot{M} = -4 \times 10^{-13} \eta \frac{LR}{M} M_{\odot} \text{ yr}^{-1},$$

where L , R , and M are all in solar units.

- (b) Assuming (incorrectly) that L , R , and η do not change with time, derive an expression for the mass of the star as a function of time. Let $M = M_0$ when the mass loss phase begins.
- (c) Using $L = 7000 L_{\odot}$, $R = 310 R_{\odot}$, $M_0 = 1 M_{\odot}$, and $\eta = 1$, make a graph of the star's mass as a function of time.
- (d) How long would it take for a star with an initial mass of $1 M_{\odot}$ to be reduced to the mass of the degenerate carbon–oxygen core ($0.6 M_{\odot}$)?
- 13.8** The Helix nebula is a planetary nebula with an angular diameter of $16'$ that is located approximately 213 pc from Earth.
- (a) Calculate the diameter of the nebula.
- (b) Assuming that the nebula is expanding away from the central star at a constant velocity of 20 km s^{-1} , estimate its age.
- 13.9** An old version of stellar evolution, popular at the beginning of the twentieth century, maintained that stars begin their lives as large, cool spheres of gas, like the giant stars on the H–R diagram. They then contract and heat up under the pull of their own gravity to become hot, bright blue O stars. For the remainder of their lives they lose energy, becoming dimmer and redder with age. As they slowly move down the main sequence, they eventually end up as cool, dim red M stars. Explain how observations of stellar clusters, plotted on an H–R diagram, contradict this idea.
- 13.10 (a)** Using data available in Tables 12.1 and 13.1, compare the pre-main sequence evolutionary time of a $0.8 M_{\odot}$ star with the lifetime on the main sequence for a $15 M_{\odot}$ star. How does this information help to explain the appearance of a color–magnitude diagram such as Fig. 13.18?
- (b) Estimate the mass of a star that would have a main-sequence lifetime comparable to the pre-main-sequence evolutionary time of a $0.8 M_{\odot}$ star.
- 13.11 (a)** The age of the universe is 13.7 Gyr. Compare this value to the main-sequence lifetime of a $0.8 M_{\odot}$ star. Why isn't it useful to compute the detailed post-main-sequence evolution of stars with masses much lower than the mass of the Sun?
- (b) Would you expect to find globular clusters with main-sequence turn-off points below $0.8 M_{\odot}$? Explain your answer.
- 13.12 (a)** Show that $\log_{10} (L_V/L_B) + \text{constant}$ is, to within a multiplicative constant, equivalent to the color index, $B - V$.
- (b) Estimating best-fit curves through the data given in Fig. 13.20, trace the two color–magnitude diagrams, placing them on a single graph. Note that the abscissas have been normalized so that the lowest-luminosity stars of both clusters are located at the same positions on their respective diagrams.
- (c) Given that 47 Tuc is relatively metal-rich for a globular cluster ($Z/Z_{\odot} = 0.17$, where $Z_{\odot}$ is the solar value) and M15 is metal-poor ($Z/Z_{\odot} = 0.0060$), explain the difference in colors between the two clusters. *Hint:* You may wish to refer back to the discussion in Example 9.5.4 (Section 9.5).

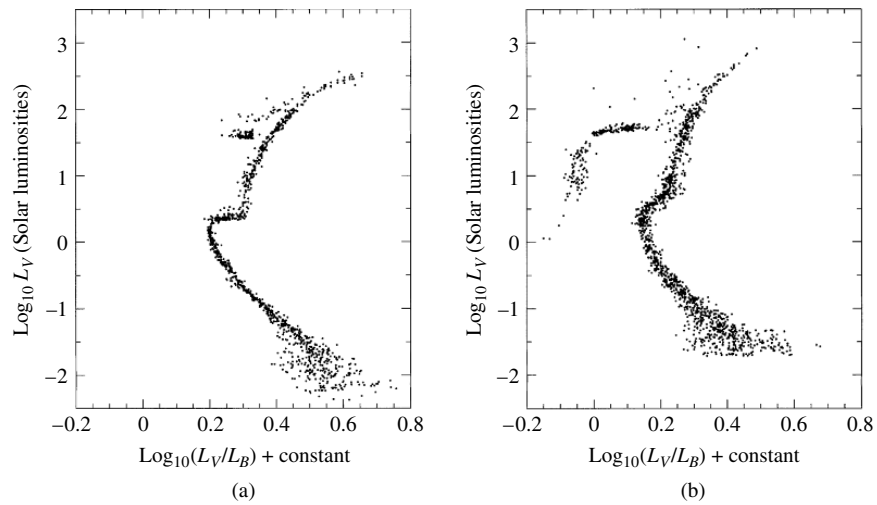


FIGURE 13.20 (a) A color–magnitude diagram for 47 Tuc, a relatively metal-rich globular cluster with $Z/Z_{\odot} = 0.17$. (Data from Hesser et al., *Publ. Astron. Soc. Pac.*, 99, 739, 1987; figure courtesy of William E. Harris.) (b) A color–magnitude diagram for M15, a metal-poor globular cluster with $Z/Z_{\odot} = 0.0060$. (Data from Durrell and Harris, *Astron. J.*, 105, 1420, 1993; figure courtesy of William E. Harris.)

13.13 Using the technique of main-sequence fitting, estimate the distance to M3; refer to Figs. 13.17 and 13.19.

CHAPTER

14

Stellar Pulsation

- 14.1 *Observations of Pulsating Stars*
- 14.2 *The Physics of Stellar Pulsation*
- 14.3 *Modeling Stellar Pulsation*
- 14.4 *Nonradial Stellar Pulsation*
- 14.5 *Helioseismology and Asteroseismology*

14.1 ■ OBSERVATIONS OF PULSATING STARS

In August of 1595, a Lutheran pastor and amateur astronomer named David Fabricius (1564–1617) observed the star α Ceti. As he watched over a period of months, the brightness of this second-magnitude star in the constellation Cetus (the Sea Monster) slowly faded. By October, the star had vanished from the sky. Several more months passed as the star eventually recovered and returned to its former brilliance. In honor of this miraculous event, α Ceti was named Mira, meaning “wonderful.”

Mira continued its rhythmic dimming and brightening, and by 1660 the 11-month period of its cycle was established. The regular changes in brightness were mistakenly attributed to dark “blotches” on the surface of a rotating star. Supposedly, Mira would appear fainter when these dark areas were turned toward Earth.

Figure 14.1 shows the *light curve* of Mira for a 51-year interval. Today astronomers recognize that the changes in Mira’s brightness are due not to dark spots on its surface but to the fact that Mira is a **pulsating star**, a star that dims and brightens as its surface expands and contracts. Mira is the prototype of the **long-period variables**, stars that have somewhat irregular light curves and pulsation periods between 100 and 700 days.

Nearly two centuries elapsed before another pulsating star was discovered. In 1784 John Goodricke (1764–1786) of York, England, found that the brightness of the star δ Cephei varies regularly with a period of 5 days, 8 hours, 48 minutes. This discovery cost Goodricke his life; he contracted pneumonia while observing δ Cephei and died at the age of 21. The light curve of δ Cephei, shown in Fig. 14.2, is less spectacular than that of α Ceti. It varies by less than one magnitude in brightness and never fades from view. Nevertheless, pulsating stars similar to δ Cephei, called **classical Cepheids**, are vitally important to astronomy.

The Period–Luminosity Relation

By 2005, nearly 40,000 pulsating stars had been cataloged by astronomers. One woman, Henrietta Swan Leavitt (1868–1921; see Fig. 14.3), discovered more than 5% of these stars

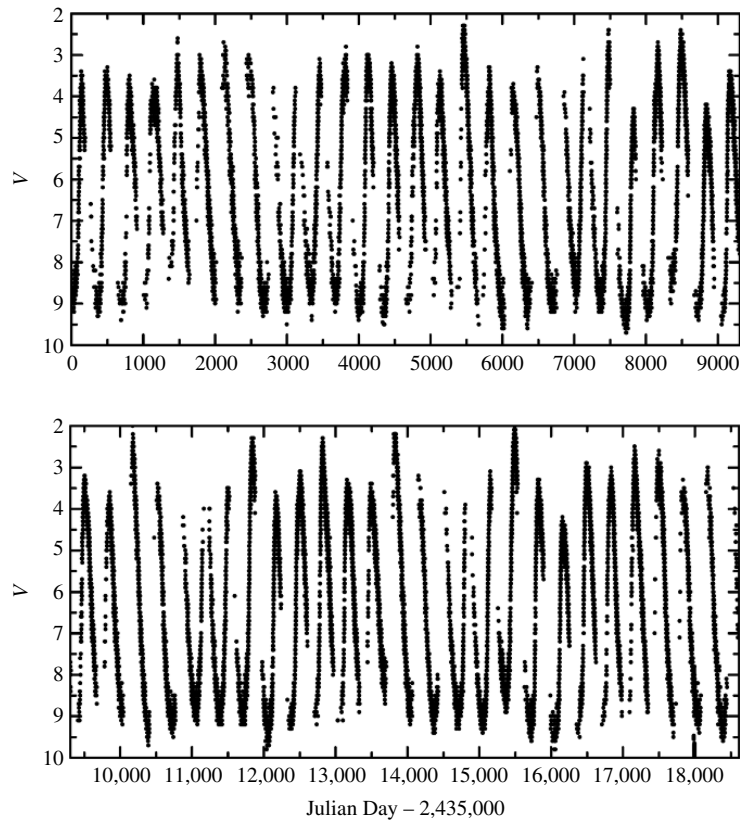


FIGURE 14.1 The light curve of Mira from September 14, 1954 (JD 2,435,000) through September 2005. Recall that magnitudes dimmer than 6 are undetectable to the unaided eye. (We acknowledge with thanks the variable-star observations from the AAVSO International Database contributed by observers worldwide.)

while working as a “computer” for Edward Charles Pickering (1846–1919) at Harvard University. Her tedious task was to compare two photographs of the same field of stars taken at different times and detect any star that varied in brightness. Eventually she discovered 2400 classical Cepheids with periods between 1 and 50 days, most of them located in the Small Magellanic Cloud (SMC). Leavitt took advantage of this opportunity to investigate the nature of the classical Cepheids in the SMC. Noticing that the more luminous Cepheids took longer to go through their pulsation cycles, she plotted the apparent magnitudes of these SMC stars against their pulsation periods. The resulting graph, shown in Fig. 14.4, demonstrated that the apparent magnitudes of classical Cepheids are closely correlated with their periods, with an uncertainty of only $\Delta m \approx \pm 0.5$ at a given period.

Because all of the stars in the Small Magellanic Cloud are roughly the same distance from us (about 61 kpc), the differences in their apparent magnitudes must be the same as the differences in their absolute magnitudes [cf. Eq. (3.6) for the distance modulus]. Thus the

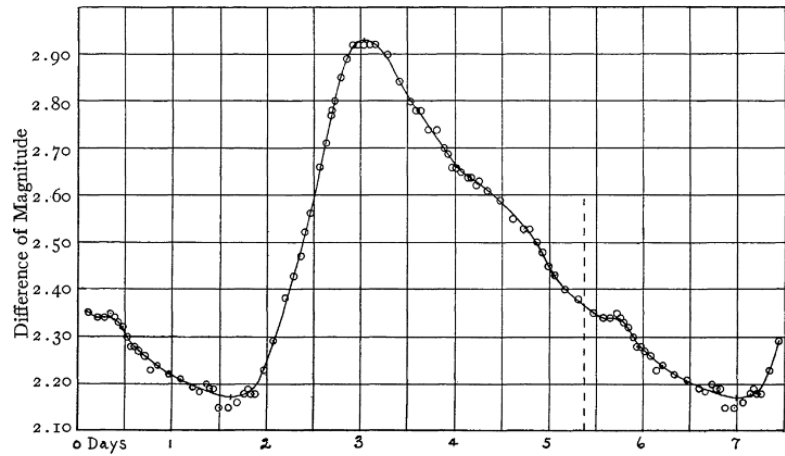


FIGURE 14.2 The light curve of δ Cephei. Its pulsation period is 5.37 days. (Figure from Stebbins, Joel, *Ap. J.*, 27, 188, 1908.)



FIGURE 14.3 Henrietta Swan Leavitt (1868–1921). (Courtesy of Harvard College Observatory.)

observed differences in these stars' apparent brightnesses must reflect intrinsic differences in their luminosities. Astronomers were excited at the prospect of determining the absolute magnitude or luminosity of a distant Cepheid simply by timing its pulsation, because knowing both a star's apparent and absolute magnitudes allows the distance of the star to be easily determined from the distance modulus, Eq. (3.6). This would permit the measurement of large distances in the universe, far beyond the limited range of parallax techniques. The only stumbling block was the calibration of Leavitt's relation. An independent distance to a

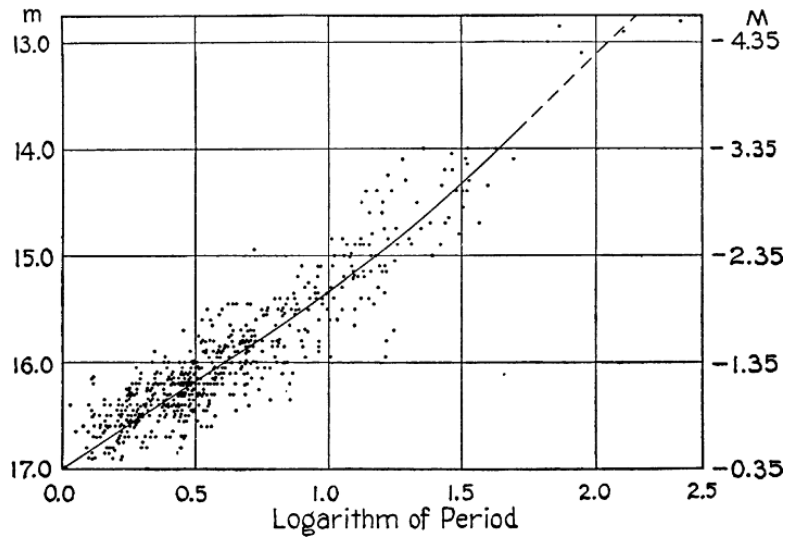


FIGURE 14.4 Classical Cepheids in the Small Magellanic Cloud, with the period in units of days. (Figure from Shapley, *Galaxies*, Harvard University Press, Cambridge, MA, 1961.)

single Cepheid had to be obtained to measure its absolute magnitude and luminosity. Once this difficult chore was accomplished, the resulting **period–luminosity relation** could be used to measure the distance to any Cepheid.

The nearest classical Cepheid is Polaris, some 200 pc away. In the early twentieth century, this distance was too great to be reliably measured by stellar parallax. However, in 1913, Ejnar Hertzsprung succeeded in using the longer baseline provided by the Sun’s motion through space, together with statistical methods, to find the distances to Cepheids having a specified period. (The measurement of the absolute magnitude of a Cepheid is also complicated by the dimming effect of interstellar extinction; recall Eq. 12.1.)

The calibrated period–luminosity relation depicted in Fig. 14.5 for the V band is described by

$$M_{(V)} = -2.81 \log_{10} P_d - 1.43, \quad (14.1)$$

where $M_{(V)}$ is the average absolute V magnitude and P_d is the pulsation period in units of days. In terms of the average luminosity of the star, the relation is given by

$$\log_{10} \frac{\langle L \rangle}{L_{\odot}} = 1.15 \log_{10} P_d + 2.47. \quad (14.2)$$

Astronomers can substantially decrease the scatter in the period–luminosity relation by making observations in infrared wavelengths where interstellar extinction is less of a problem. One such fit, made using magnitudes measured in the infrared H band (centered at $1.654 \mu\text{m}$), is illustrated in Fig. 14.6(a). The data are for 92 Cepheids in the Large

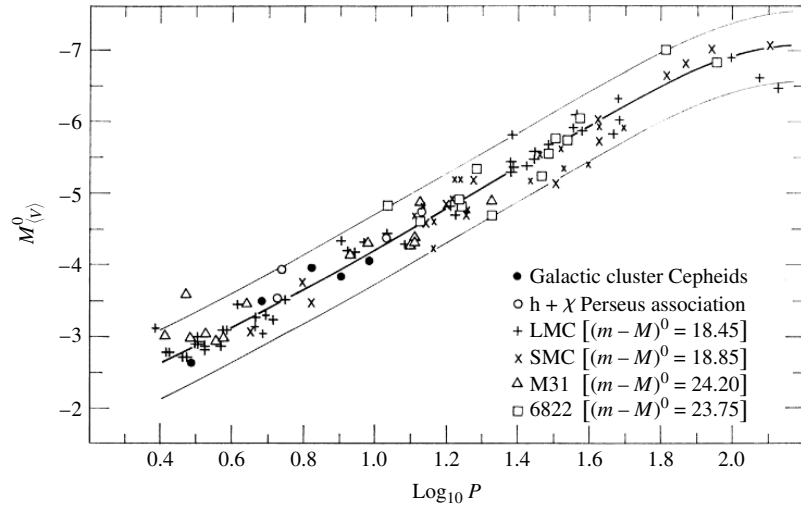


FIGURE 14.5 The period–luminosity relation for classical Cepheids. (Figure adapted from Sandage and Tammann, *Ap. J.*, 151, 531, 1968.)

Magellanic Cloud. The infrared period–luminosity fit is given by

$$H = -3.234 \log_{10} P_d + 16.079. \quad (14.3)$$

The scatter can be further reduced by adding a color term to the fit. Using the infrared color index $J - K_s$, Fig. 14.6(b) shows that the fit is indeed somewhat tighter (J and K_s are centered at $1.215 \mu\text{m}$ and $2.157 \mu\text{m}$, respectively). The fit for this **period–luminosity–color relation** is given by

$$H = -3.428 \log_{10} P_d + 1.54(J - K_s) + 15.637. \quad (14.4)$$

Classical Cepheids provide astronomy with its third dimension and supply the foundation for the measurement of extragalactic distances. Because Cepheids are supergiant stars (luminosity class Ib), about fifty times the Sun’s size and thousands of times more luminous, they can be seen over intergalactic distances. They serve as “standard candles,” beacons scattered throughout the night sky that serve as mileposts for astronomical surveys of the universe.

The Pulsation Hypothesis for Brightness Variations

The important use of Cepheids as cosmic distance indicators does not require an understanding of the physical reasons for their light variations. In fact, the observed changes in brightness were once thought to be caused by tidal effects in the atmospheres of binary stars. However, in 1914 the American astronomer Harlow Shapley (1885–1972) argued that the binary theory was fatally flawed because the size of the star would exceed the size of the

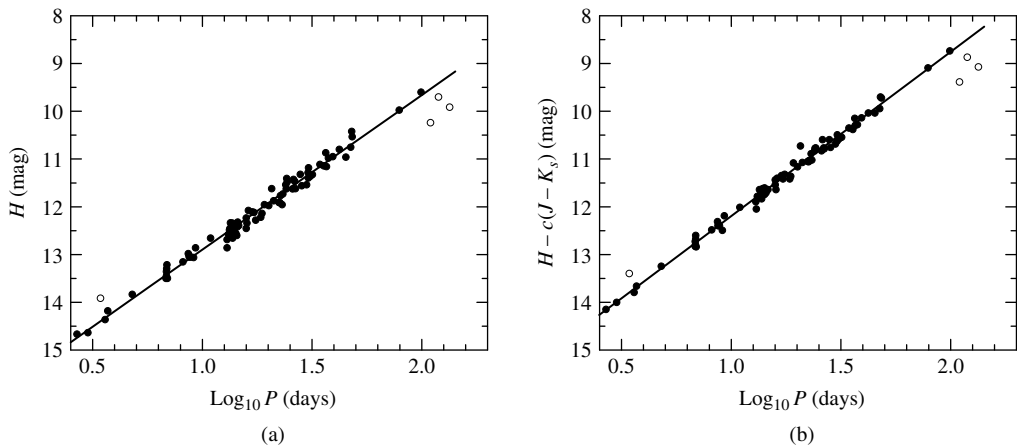


FIGURE 14.6 (a) The infrared period–luminosity relation for 92 Cepheids in the Large Magellanic Cloud. The infrared H band was used for the observations. The open circles represent four Cepheids that were excluded from the least-squares linear fit. (b) The period–luminosity–color relation for the same Cepheids. (Data from Persson, S. E., et al., *Astron. J.*, 128, 2239, 2004.)

orbit for some variables. Shapley advanced an alternative idea: that the observed variations in the brightness and temperature of classical Cepheids were caused by the radial pulsation of single stars. He proposed that these stars were rhythmically “breathing” in and out, becoming alternately brighter and dimmer in the process. Four years later Sir Arthur Stanley Eddington provided a firm theoretical framework for the pulsation hypothesis, which received strong support from the observed correlations among the variations in brightness, temperature, and surface velocity throughout the pulsation cycle. Figure 14.7 shows the measured changes in magnitude, temperature, radius, and surface velocity for δ Cephei. The change in brightness is primarily due to the roughly 1000 K variation in δ Cephei’s surface temperature; the accompanying change in size makes a lesser contribution to the luminosity. Although the total excursion of δ Cephei’s surface from its equilibrium radius is large in absolute terms (a bit more than the diameter of the Sun), it is still only about 5% to 10% of the size of this supergiant star. The spectral type of δ Cephei changes continuously throughout the cycle, varying between F5 (hottest) and G2 (coolest). A careful examination of Fig. 14.7 reveals that the magnitude and surface velocity curves are nearly identical in shape. Thus, the star is brightest when its surface is expanding outward most rapidly, *after* it has passed through its minimum radius. Later in this chapter we will see that the explanation of this **phase lag** of maximum luminosity behind minimum radius has its origin in the mechanism that maintains the oscillations.

The Instability Strip

The Milky Way Galaxy is estimated to contain several million pulsating stars. Considering that our Galaxy consists of several hundred billion stars, this implies that stellar pulsation must be a transient phenomenon. The positions of the pulsating variables on the H – R diagram (see Figs. 14.8 and 8.16) confirm this conclusion. Rather than being located on the main sequence, where stars spend most of their lives, the majority of pulsating stars occupy

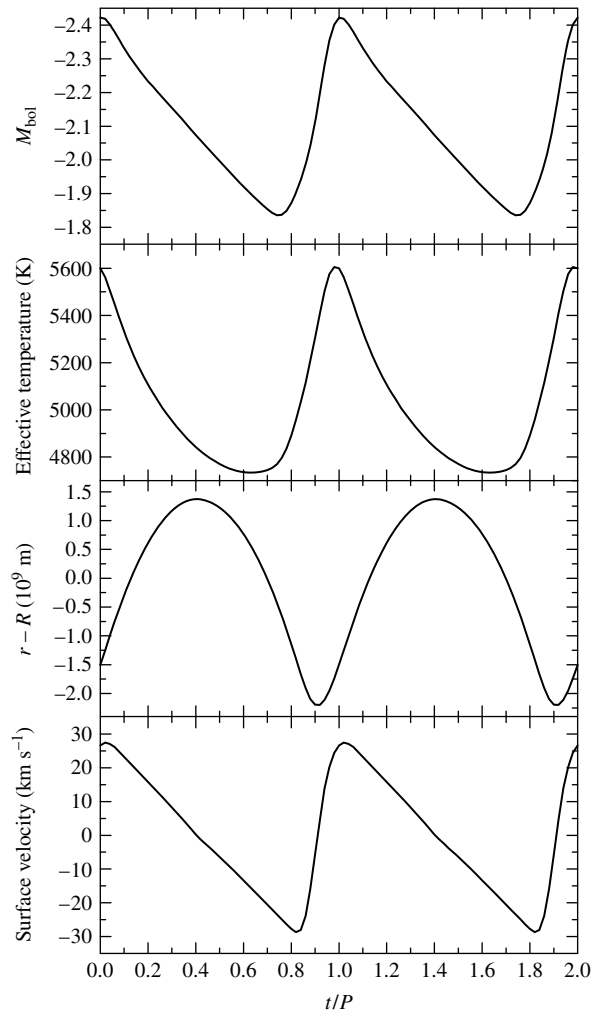


FIGURE 14.7 Observed pulsation properties of δ Cephei, a typical classical Cepheid. (Data from Schwarzschild, *Harvard College Observatory Circular*, 431, 1938.)

a narrow (about 600–1100 K wide), nearly vertical **instability strip** on the right-hand side of the H–R diagram. Theoretical evolutionary tracks for stars of various masses are also shown in Fig. 14.8. As stars evolve along these tracks, they begin to pulsate as they enter the instability strip and cease their oscillations upon leaving. Of course, evolutionary timescales are far too long for us to observe the onset and cessation of a single star’s oscillations, but several stars have been caught in the final phase of their pulsational history.

Some Classes of Pulsating Stars

Astronomers have divided pulsating stars into several classes. Some of these are listed in Table 14.1. The W Virginis stars are metal-deficient (Population II) Cepheids and are

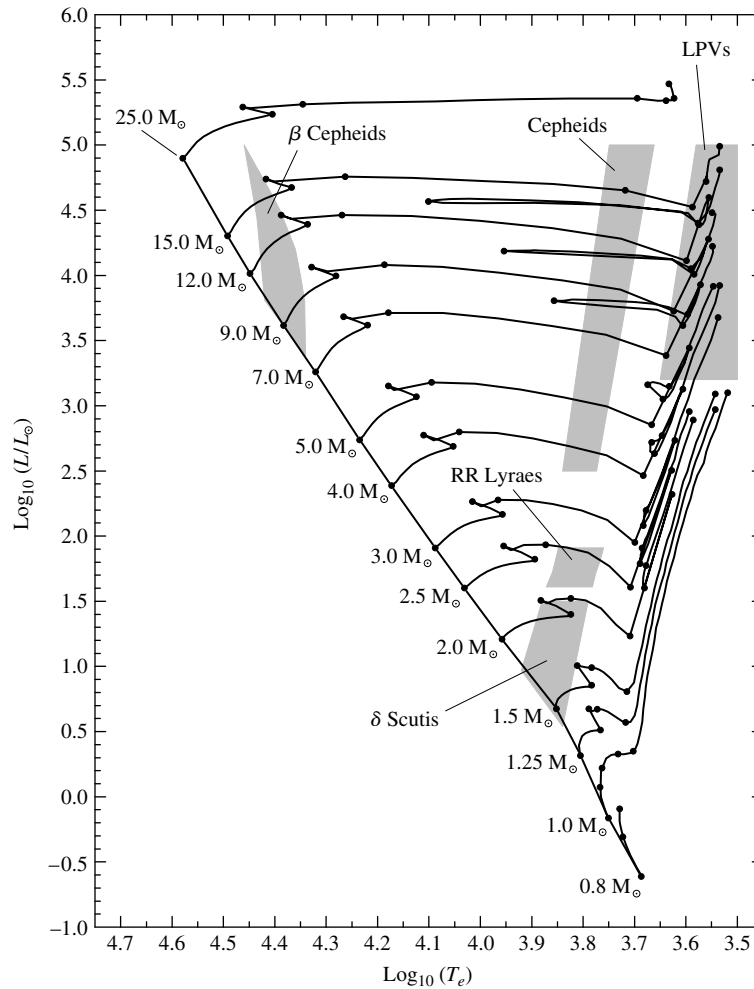


FIGURE 14.8 Pulsating stars on the H–R diagram. (Data for the evolutionary tracks from Schaller, et al., *Astron. Astrophys. Suppl.*, 96, 269, 1992.)

about four times less luminous than classical Cepheids with the same period. Their period–luminosity relation is thus lower than and parallel to the one shown for the classical Cepheids in Fig. 14.5. RR Lyrae stars, also Population II, are horizontal-branch stars found in globular clusters. Because all RR Lyrae stars have nearly the same luminosity, they are also useful yardsticks for distance measurements. The δ Scuti variables are evolved F stars found near the main sequence of the H–R diagram. They exhibit both radial and nonradial oscillations; the latter is a more complicated motion that will be discussed in Section 14.4. Below the main sequence (not shown in Fig. 14.8; however, see Fig. 16.4) are the pulsating white dwarfs, called ZZ Ceti stars.

All of the types of stars listed thus far lie within the instability strip, and they share a common mechanism that drives the oscillations. The long-period variables such as Mira

TABLE 14.1 Pulsating Stars. (Adopted from Cox, *The Theory of Stellar Pulsation*, Princeton University Press, Princeton, NJ, 1980.)

Type	Range of Periods	Population Type	Radial or Nonradial
Long-Period Variables	100–700 days	I,II	R
Classical Cepheids	1–50 days	I	R
W Virginis stars	2–45 days	II	R
RR Lyrae stars	1.5–24 hours	II	R
δ Scuti stars	1–3 hours	I	R,NR
β Cephei stars	3–7 hours	I	R,NR
ZZ Ceti stars	100–1000 seconds	I	NR

and the β Cephei stars are located outside of the instability strip occupied by the classical Cepheids and RR Lyrae stars. Their unusual positions on the H–R diagram will be discussed in the next section.

14.2 ■ THE PHYSICS OF STELLAR PULSATION

Geologists and geophysicists have obtained a wealth of information about Earth’s interior from their study of the seismic waves produced by earthquakes and other sources. In the same manner, astrophysicists model the pulsational properties of stars to understand better their internal structure. By numerically calculating an evolutionary sequence of stellar models and then comparing the pulsational characteristics (periods, amplitudes, and details of the light and radial velocity curves) of the models with those actually observed, astronomers can further test their theories of stellar structure and evolution and obtain a detailed view of the interior of a star.¹

The Period–Density Relation

The radial oscillations of a pulsating star are the result of sound waves resonating in the star’s interior. A rough estimate of the pulsation period,² Π , may be easily obtained by considering how long it would take a sound wave to cross the diameter of a model star of radius R and constant density ρ . The adiabatic sound speed is given by Eq. (10.84),

$$v_s = \sqrt{\frac{\gamma P}{\rho}}.$$

The pressure may be found from Eq. (10.6) for hydrostatic equilibrium, using the

¹Several other ways of testing the ideas of stellar structure and evolution were discussed in Chapter 13.

²Throughout the following discussion, Π will be used to designate the pulsation period so that it is not confused with the pressure, P . Π is commonly used for the pulsation period in stellar pulsation theory studies. (T , another symbol commonly used for period, would lead to confusion with temperature.)

(unrealistic) assumption of constant density. Thus

$$\frac{dP}{dr} = -\frac{GM_r \rho}{r^2} = -\frac{G\left(\frac{4}{3}\pi r^3 \rho\right) \rho}{r^2} = -\frac{4}{3}\pi G \rho^2 r.$$

This is readily integrated using the boundary condition that $P = 0$ at the surface to obtain the pressure as a function of r ,

$$P(r) = \frac{2}{3}\pi G \rho^2 (R^2 - r^2). \quad (14.5)$$

Thus the pulsation period is roughly

$$\Pi \approx 2 \int_0^R \frac{dr}{v_s} \approx 2 \int_0^R \frac{dr}{\sqrt{\frac{2}{3}\gamma \pi G \rho (R^2 - r^2)}},$$

or

$$\boxed{\Pi \approx \sqrt{\frac{3\pi}{2\gamma G \rho}}}. \quad (14.6)$$

Qualitatively, this shows that the pulsation period of a star is inversely proportional to the square root of its mean density. Referring to Fig. 14.8 and Table 14.1, this **period–mean density relation** explains why the pulsation period decreases as we move down the instability strip from the very tenuous supergiants to the very dense white dwarfs.³ The tight period–luminosity relation discovered by Leavitt exists because the instability strip is roughly parallel to the luminosity axis of the H–R diagram (the finite width of the instability strip is reflected in the ± 0.5 magnitude uncertainty in the period–luminosity relation). The quantitative agreement of Eq. (14.6) with the observed periods of Cepheids is not too bad, considering its crude derivation. If we take $M = 5 M_\odot$ and $R = 50 R_\odot$ for a typical Cepheid, then $\Pi \approx 10$ days. This falls nicely within the range of periods measured for the classical Cepheids.

Radial Modes of Pulsation

The sound waves involved in the **radial modes** of stellar pulsation are essentially *standing waves*, similar to the standing waves that occur in an organ pipe that is open at one end; see Fig. 14.9. Both the star and the organ pipe can sustain several modes of oscillation. The standing wave for each mode has a *node* at one end (the star’s center, the pipe’s closed end), where the gases do not move, and an *antinode* at the other end (the star’s surface, the pipe’s open end). For the **fundamental mode**, the gases move in the same direction at every point in the star or pipe. There is a single node between the center and the surface for the **first overtone** mode,⁴ with the gases moving in opposite directions on either side of

³Pulsating white dwarfs exhibit nonradial oscillations, and their periods are longer than predicted by the period–mean density relation.

⁴Some texts use the unfortunate term *first harmonic* for the first overtone.

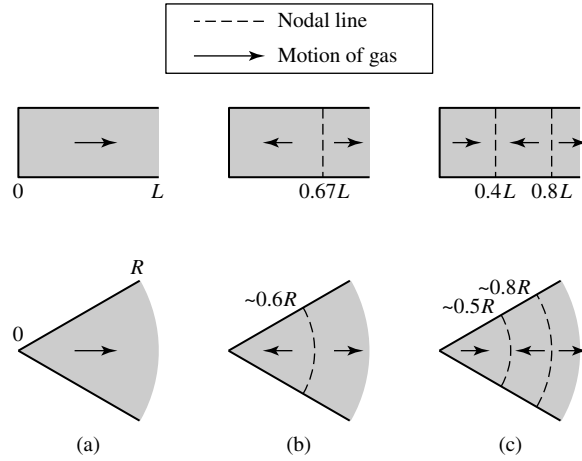


FIGURE 14.9 Standing sound waves in an organ pipe and in a star for (a) the fundamental mode, (b) the first overtone, and (c) the second overtone.

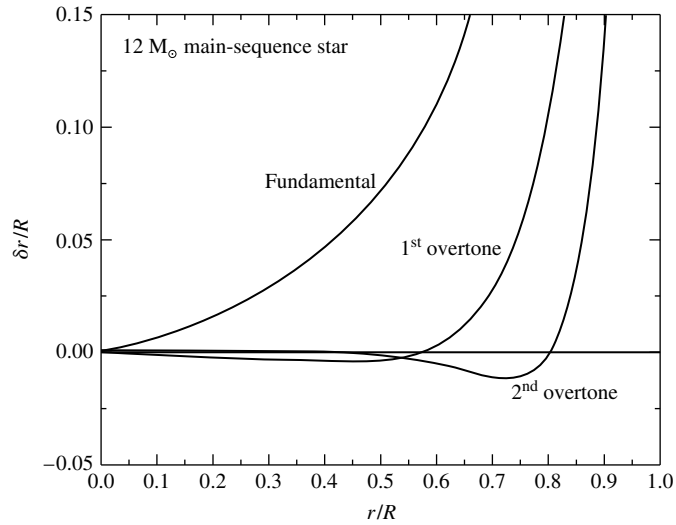


FIGURE 14.10 Radial modes for a pulsating star. The waveform for each mode has been arbitrarily scaled so that $\delta r/R = 1$ at the surface of the star. Actually, the maximum surface ratio of $\delta r/R$ is approximately 0.05 to 0.10 for a classical Cepheid.

the node, and two nodes for the **second overtone** mode. Figure 14.10 shows the fractional displacement, $\delta r/R$, of the stellar material from its equilibrium position for several radial modes of a $12 M_{\odot}$ main-sequence star model. Note that $\delta r/R$ has been arbitrarily scaled to unity at the stellar surface.

For radial modes, the motion of the stellar material occurs primarily in the surface regions, but there is some oscillation deep inside the star. This effect is most prominent for

the fundamental mode, where non-negligible amplitudes exist. For the stellar model used in Fig. 14.10, at $r = 0.5R$, $\delta r/R$ is about 7% of its surface value. For the first overtone at the same location, $\delta r/R$ is less than 1% of its surface value and is in the opposite direction, and for the second overtone, the oscillation is nearly zero amplitude ($r = 0.5R$ is close to a node for the second overtone).

The vast majority of the classical Cepheids and W Virginis stars pulsate in the fundamental mode. The RR Lyrae variables pulsate in either the fundamental or the first overtone mode, with a few oscillating in both modes simultaneously. The long-period variables, such as Mira, may also oscillate in either the fundamental mode or the first overtone, although this is still not entirely clear.

Eddington's Thermodynamic Heat Engine

To explain the mechanism that powers these standing sound waves, Eddington proposed that pulsating stars are thermodynamic heat engines. The gases comprising the layers of the star do $P dV$ work as they expand and contract throughout the pulsation cycle. If the integral $\oint P dV > 0$ for the cycle, a layer does net positive work on its surroundings and contributes to driving the oscillations; if $\oint P dV < 0$, the net work done by the layer is negative and tends to dampen the oscillations. Figures 14.11 and 14.12 show P - V diagrams for a driving layer and a damping layer, respectively, in a numerical calculation of the oscillation of an RR Lyrae star. If the total work (found by adding up the contributions of all the layers of the star) is positive, the oscillations will grow in amplitude. The oscillations will decay if the total work is negative. These changes in the pulsation amplitude continue until an equilibrium value is reached, when the total work done by all the layers is zero.

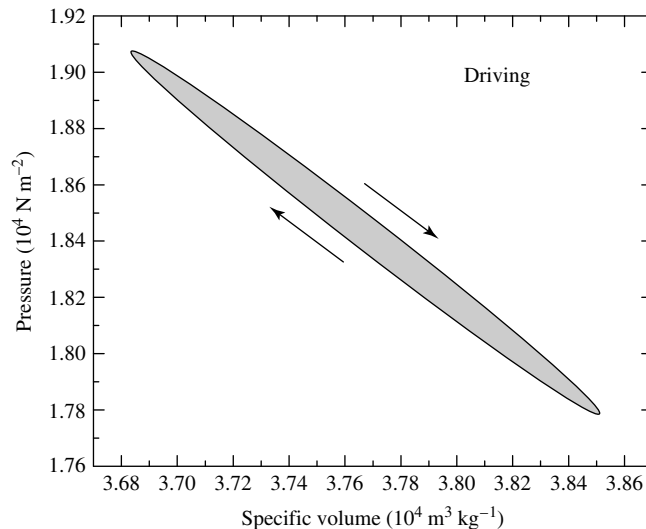


FIGURE 14.11 P - V diagram for a driving layer of an RR Lyrae star model. You may recall the analogous use of P - V diagrams in discussing heat engines in introductory physics courses. A clockwise path in a P - V diagram corresponds with net driving.

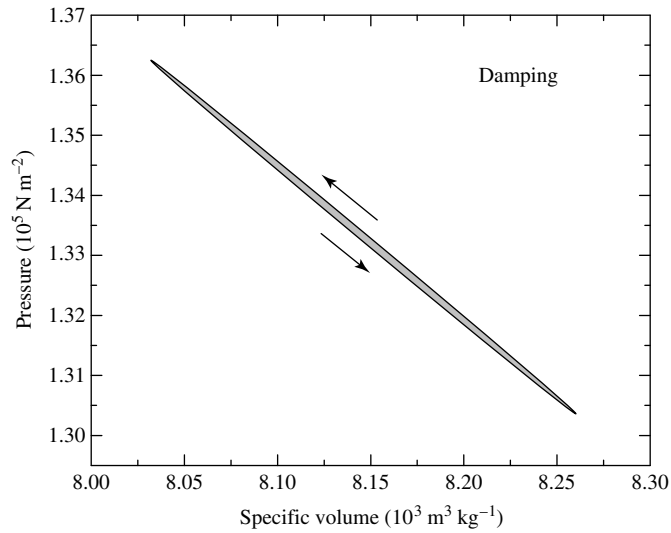


FIGURE 14.12 P - V diagram for a damping layer of an RR Lyrae star model. A counterclockwise path in a P - V diagram corresponds to net damping.

As for any heat engine, the net work done by each layer of the star during one cycle is the difference between the heat flowing into the gas and the heat leaving the gas. For driving, the heat must enter the layer during the high-temperature part of the cycle and leave during the low-temperature part. Just as the spark plug of an automobile engine fires at the end of the compression stroke, the driving layers of a pulsating star must absorb heat around the time of their maximum compression. In this case the maximum pressure will occur *after* maximum compression, and the oscillations will be amplified.

The Nuclear ϵ Mechanism

In what region of the star can this driving take place? An obvious possibility was first considered by Eddington: When the center of the star is compressed, its temperature and density rise, increasing the rate at which thermonuclear energy is generated. However, recall from Fig. 14.10 that the displacement $\delta r/R$ has a node at the center of the star. The pulsation amplitude is very small near the center. Although this energy mechanism (called the ϵ -**mechanism**) does in fact operate in the core of a star, it is usually not enough to drive the star's pulsation. However, as mentioned in Section 10.6, variations in the nuclear energy generation rate (ϵ) produce oscillations that may contribute to preventing the formation of stars with masses greater than approximately $90 M_{\odot}$.

Eddington's Valve

Eddington then suggested an alternative, a *valve mechanism*. If a layer of the star became more opaque upon compression, it could “dam up” the energy flowing toward the surface and push the surface layers upward. Then, as this expanding layer became more transparent, the trapped heat could escape and the layer would fall back down to begin the cycle anew. In Eddington's own words, “To apply this method we must make the star more heat-tight

when compressed than when expanded; in other words, *the opacity must increase with compression.*”

In most regions of the star, however, the opacity actually *decreases* with compression. Recall from Section 9.2 that for a Kramers law, the opacity κ depends on the density and temperature of the stellar material as $\kappa \propto \rho/T^{3.5}$. As the layers of a star are compressed, their density and temperature both increase. But because the opacity is more sensitive to the temperature than to the density, the opacity of the gases usually decreases upon compression. It takes special circumstances to overcome the damping effect of most stellar layers, which explains why stellar pulsation is observed for only one of every 10^5 stars.

Opacity Effects and the κ and γ Mechanisms

The conditions responsible for exciting and maintaining the stellar oscillations were first identified by the Russian astronomer S. A. Zhevakin and then verified in detailed calculations by a German and two Americans, Rudolph Kippenhahn, Norman Baker, and John P. Cox (1926–1984). They found that the regions of a star where Eddington’s valve mechanism can successfully operate are its *partial ionization zones* (cf. Section 8.1). In these layers of the star where the gases are partially ionized, part of the work done on the gases as they are compressed produces further ionization rather than raising the temperature of the gas.⁵ With a smaller temperature rise, the increase in density with compression produces a corresponding increase in the Kramers opacity; see Fig. 14.13. Similarly, during expansion, the temperature does not decrease as much as expected since the ions now recombine with electrons and release energy. Again, the density term in the Kramers law dominates, and

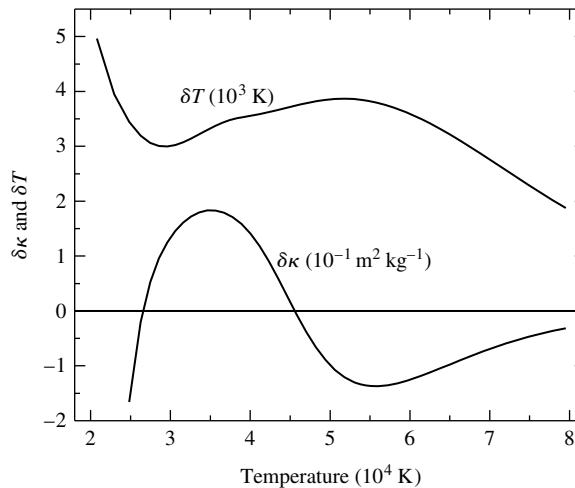


FIGURE 14.13 Variations in the temperature and opacity throughout an RR Lyrae star model at the time of maximum compression. In the He II partial ionization zone ($T \approx 40,000$ K), $\delta\kappa > 0$ and δT is reduced. These are the κ - and γ -mechanisms that drive the star’s oscillations.

⁵As discussed in Section 10.4, this causes the specific heats C_P and C_V to have larger values in a partial ionization zone.

the opacity decreases with decreasing density during the expansion. This layer of the star can thus absorb heat during compression, be pushed outward to release the heat during expansion, and fall back down again to begin another cycle. Astronomers refer to this opacity mechanism as the **κ -mechanism**.

In a partial ionization zone, the κ -mechanism is reinforced by the tendency of heat to flow into the zone during compression simply because its temperature has increased less than the adjacent stellar layers. This effect is called the **γ -mechanism**, after the smaller ratio of specific heats caused by the increased values of C_P and C_V . Partial ionization zones are the pistons that drive the oscillations of stars; they modulate the flow of energy through the layers of the star and are the direct cause of stellar pulsation.

The Hydrogen and Helium Partial Ionization Zones

In most stars there are two main ionization zones. The first is a broad zone where both the ionization of neutral hydrogen ($\text{H I} \rightarrow \text{H II}$) and the first ionization of helium ($\text{He I} \rightarrow \text{He II}$) occur in layers with a characteristic temperature of 1 to 1.5×10^4 K. These layers are collectively referred to as the **hydrogen partial ionization zone**. The second, deeper zone involves the second ionization of helium ($\text{He II} \rightarrow \text{He III}$), which occurs at a characteristic temperature of 4×10^4 K and is called the **He II partial ionization zone**.

The location of these ionization zones within the star determines its pulsational properties. As shown in Fig. 14.14, if the star is too hot (7500 K), the ionization zones will be located very near the surface. At this position, the density is quite low, and there is not enough mass available to drive the oscillations effectively. This accounts for the hot **blue edge** of the instability strip on the H–R diagram. In a cooler star (6500 K), the characteristic

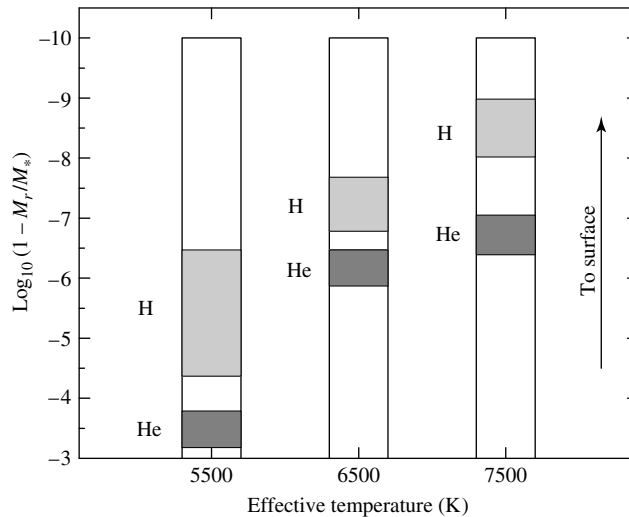


FIGURE 14.14 Hydrogen and helium ionization zones in stars of different temperatures. For each point in the star, the vertical axis displays the logarithm of the fraction of the star's mass that lies *above* that point.

temperatures of the ionization zones are found deeper in the star. There is more mass for the ionization zone “piston” to push around, and the first overtone mode may be excited. (Whether a mode is actually excited depends on whether the positive work generated within the ionization zones is sufficient to overcome the damping due to negative work of the other layers of the star.) In a still cooler star (5500 K), the ionization zones occur deep enough to drive the fundamental mode of pulsation. However, if a star’s surface temperature is too low, the onset of efficient convection in its outer layers may dampen the oscillations. Because the transport of energy by convection is more effective when the star is compressed, the convecting stellar material may lose heat at minimum radius. This could overcome the damming up of heat by the ionization zones—and so quench the pulsation of the star. The cool **red edge** of the instability strip is the result of the damping effect of convection.⁶

Detailed numerical calculations of the pulsation of model stars produce an instability strip that is in good agreement with its observed location on the H–R diagram. These computations show that it is the He II partial ionization zone that is primarily responsible for driving the oscillations of stars within the instability strip. If the effect of the helium ionization zone is artificially removed, the model stars will not pulsate.

The hydrogen ionization zone plays a more subtle role. As a star pulsates, the hydrogen ionization zone moves toward or away from the surface as the zone expands and contracts in response to the changing temperature of the stellar gases. It happens that the star is brightest when the *least mass* lies between the hydrogen ionization zone and the surface. As a star oscillates, the location of an ionization zone changes with respect to both its radial position, r , and its mass interior to r , M_r . The luminosity incident on the *bottom* of the hydrogen ionization zone is indeed a maximum at minimum radius, but this merely propels the zone outward (through mass) most rapidly at that instant. The emergent luminosity is thus greatest *after* minimum radius, when the zone is nearest the surface. This delaying action of the hydrogen partial ionization zone produces the phase lag observed for classical Cepheids and RR Lyrae stars.

The mechanisms responsible for the pulsation of stars outside the instability strip are not always as well understood. The long-period variables are red supergiants (AGB stars) with huge, diffuse convective envelopes surrounding a compact core. Their spectra are dominated by molecular absorption lines and emission lines that reveal the existence of atmospheric shock waves and significant mass loss. While we understand that the *hydrogen* partial ionization zone drives the pulsation of a long-period variable star, many details remain to be explained, such as how its oscillations interact with its outer atmosphere.⁷

β Cephei Stars and the Iron Opacity “Bump”

The β Cephei stars pose another interesting challenge. Being situated in the upper left-hand side of the H–R diagram, these stars are very hot and luminous. β Cepheids are early B stars with effective temperatures in the range of 20,000 to 30,000 K and typically with luminosity classes of III, IV, and V. Given their high effective temperatures, hydrogen is completely

⁶Much work remains to be done on the effect of convection on stellar pulsation, although some results have been obtained for RR Lyrae and ZZ Ceti stars. Progress has been hampered by the present lack of a fundamental theory of time-dependent convection.

⁷In Section 16.2, we will see that the ZZ Ceti stars are also driven by the hydrogen partial ionization zone.

ionized, and the helium ionization zone is too near the surface to effectively drive pulsations in these stars. After years of investigation it was realized that the κ and γ mechanisms are still active in β Cephei stars, but the element responsible for the driving is iron. Although the abundance of iron is low in all stars (recall, for example, Table 9.2), the large number of absorption lines in the spectrum of iron implies that iron contributes significantly to stellar opacities at temperatures near 100,000 K. This effect can be seen in the “iron bump” above 100,000 K in the plot of opacity vs. temperature shown in Fig. 9.10. The depth of this iron ionization region is sufficient to produce net positive pulsational driving in these stars.

14.3 ■ MODELING STELLAR PULSATION

In Section 10.5, the construction of a stellar model in hydrostatic equilibrium was described. The star was considered to be divided into a number of concentric mass shells. The differential equations of static stellar structure were then converted into difference equations and applied to each mass shell, and the system of equations was solved on a computer subject to certain boundary conditions at the center and surface of the stellar model.

Nonlinear Hydrodynamic Models

Because a pulsating star is not in hydrostatic equilibrium, the stellar structure equations collected at the beginning of Section 10.5 cannot be used in their present form. Instead, a more general set of equations is employed that takes the oscillation of the mass shells into account. For example, Newton’s second law (Eq. 10.5),

$$\rho \frac{d^2 r}{dt^2} = -G \frac{M_r \rho}{r^2} - \frac{dP}{dr}, \quad (14.7)$$

must be used instead of Eq. (10.6) for hydrostatic equilibrium. Once the differential equations describing the nonequilibrium mechanical and thermal behavior of a star have been assembled, along with the appropriate constitutive relations, they may be replaced by difference equations as described in Section 10.5 and solved numerically. In essence, the model star is mathematically displaced from its equilibrium configuration and then “released” to begin its oscillation. The mass shells expand and contract, pushing against each other as they move. If conditions are right, the ionization zones in the model star will drive the oscillations, and the pulsation amplitude will slowly increase; otherwise the amplitude will decay away. Computer programs that carry out these calculations have been quite successful at modeling the details observed in the light and radial velocity curves of Cepheid variables.

The main advantage of the preceding approach is that it is a **nonlinear** calculation, capable in principle of modeling the complexities of large pulsation amplitudes and reproducing the nonsinusoidal shape of actual light curves. One disadvantage lies in the computer resources required: This process requires a significant amount of CPU time and memory. Many (sometimes thousands of) oscillations must be calculated before the model settles down into a well-behaved periodic motion, and even more periods may be required for the model to reach its *limit cycle*, when the pulsation amplitude has reached its final value. In fact, in some cases the computer simulations of certain classes of pulsating stars may never

attain a truly periodic solution but exhibit chaotic behavior instead, as observed in some real stars.

A second disadvantage of nonlinear calculations lies in the challenges involved in accurately converging models at each time step. Numerical instabilities in the nonlinear equations can cause calculations to misbehave and lead to unphysical solutions. This is particularly true when theories of time-dependent convection are required for red giants and supergiant stars.

Linearizing the Hydrodynamic Equations

An alternative to the nonlinear approach is to **linearize** the differential equations by considering only small-amplitude oscillations. This is done by writing every variable in the differential equations as an equilibrium value (found in the static model of the star) plus a small change due to the pulsation. For example, the pressure P would be written as $P = P_0 + \delta P$, where P_0 is the value of the pressure in a mass shell of the equilibrium model, and δP is the small change in pressure that occurs as that mass shell moves in the oscillating model star. Thus δP is a function of time, but P_0 is constant. When the variables written in this manner are inserted into the differential equations, the terms containing only equilibrium quantities cancel, and terms that involve powers of the deltas higher than the first, such as $(\delta P)^2$, may be discarded because they are negligibly small. The resulting linearized differential equations and their associated boundary conditions, also linearized, are similar to the equations for a wave on a string or in an organ pipe. Only certain standing waves with specific periods are permitted, and so the pulsation modes of the star are cleanly identified. The equations are still sufficiently complicated that a computer solution is required, but the time involved is much less than that required for a nonlinear calculation. The penalties for adopting the linearized approach are that the motion of the star is forced to be sinusoidal (as it must be for small amplitudes of oscillation), and the limiting value of the pulsation amplitude cannot be determined. Modeling the complexities of the full nonlinear behavior of the stellar model is thus sacrificed.

Example 14.3.1. In this example, we consider an unrealistic, but very instructive, model of a pulsating star called a **one-zone model**; see Fig. 14.15. It consists of a central point mass equal to the entire mass of the star, M , surrounded by a single thin, spherical shell of mass m and radius R that represents the surface layer of the star. The interior of the shell is filled with a massless gas of pressure P whose sole function is to support the shell against the gravitational pull of the central mass M . Newton's second law (Eq. 14.7) applied to this shell is

$$m \frac{d^2 R}{dt^2} = -\frac{GMm}{R^2} + 4\pi R^2 P. \quad (14.8)$$

For the equilibrium model, the left-hand side of this equation is zero, so

$$\frac{GMm}{R_0^2} = 4\pi R_0^2 P_0. \quad (14.9)$$

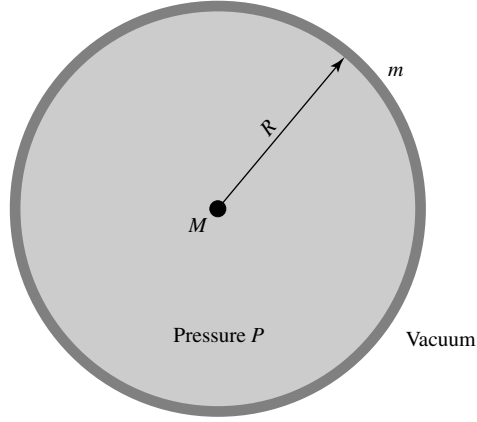


FIGURE 14.15 One-zone model of a pulsating star.

The linearization is accomplished by writing the star's radius and pressure as

$$R = R_0 + \delta R \quad \text{and} \quad P = P_0 + \delta P$$

and inserting these expressions into Eq. (14.8), giving

$$m \frac{d^2(R_0 + \delta R)}{dt^2} = -\frac{GMm}{(R_0 + \delta R)^2} + 4\pi(R_0 + \delta R)^2(P_0 + \delta P).$$

Using the first-order approximation

$$\frac{1}{(R_0 + \delta R)^2} \approx \frac{1}{R_0^2} \left(1 - 2\frac{\delta R}{R_0}\right)$$

and keeping only those terms involving the first powers of the deltas results in

$$m \frac{d^2(\delta R)}{dt^2} = -\frac{GMm}{R_0^2} + \frac{2GMm}{R_0^3} \delta R + 4\pi R_0^2 P_0 + 8\pi R_0 P_0 \delta R + 4\pi R_0^2 \delta P,$$

where $d^2 R_0 / dt^2 = 0$ has been used for the equilibrium model. The first and third terms on the right-hand side cancel (see Eq. 14.9), leaving

$$m \frac{d^2(\delta R)}{dt^2} = \frac{2GMm}{R_0^3} \delta R + 8\pi R_0 P_0 \delta R + 4\pi R_0^2 \delta P. \quad (14.10)$$

This is the linearized version of Newton's second law for our one-zone model.

To reduce the two variables δR and δP to one, we now assume that the oscillations are *adiabatic*. In this case, the pressure and volume of the model are related by the adiabatic

continued

relation $PV^\gamma = \text{constant}$, where γ is the ratio of specific heats of the gas. Since the volume of the one-zone model is just $\frac{4}{3}\pi R^3$, the adiabatic relation says that $PR^{3\gamma} = \text{constant}$. It is left as a problem to show that the linearized version of this expression is

$$\frac{\delta P}{P_0} = -3\gamma \frac{\delta R}{R_0}. \quad (14.11)$$

Using this equation, δP can be eliminated from Eq. (14.10). In addition, $8\pi R_0 P_0$ can be replaced by $2GMm/R_0^3$ through the use of Eq. (14.9). As a result, the mass m of the shell cancels, leaving the linearized equation for δR :

$$\frac{d^2(\delta R)}{dt^2} = -(3\gamma - 4) \frac{GM}{R_0^3} \delta R. \quad (14.12)$$

If $\gamma > 4/3$ (so the right-hand side of the equation is negative), this is just the familiar equation for simple harmonic motion. It has the solution $\delta R = A \sin(\omega t)$, where A is the pulsation amplitude and ω is the angular pulsation frequency. Inserting this expression for δR into Eq. (14.12) results in

$$\omega^2 = (3\gamma - 4) \frac{GM}{R_0^3}. \quad (14.13)$$

Finally, the pulsation period of the one-zone model is just $\Pi = 2\pi/\omega$, or

$$\Pi = \frac{2\pi}{\sqrt{\frac{4}{3}\pi G\rho_0(3\gamma - 4)}}, \quad (14.14)$$

where $\rho_0 = M/\frac{4}{3}\pi R_0^3$ is the average density of the equilibrium model. For an ideal monatomic gas (appropriate for hot stellar gases), $\gamma = 5/3$. Except for factors of order unity, this is the same as our earlier period estimate (Eq. 14.6) obtained by considering the time required for a sound wave to cross the diameter of a star.

In Example 14.3.1, the approximations that the pulsation of the one-zone model was *linear* and *adiabatic* were used to simplify the calculation. Note that the pulsation amplitude, A , canceled in this example. The inability to calculate the amplitude of the oscillations is an inherent drawback of the linearized approach to pulsation.

Nonlinear and Nonadiabatic Calculations

Because no heat is allowed to enter or leave the layers of a stellar model in an adiabatic analysis, the amplitude (whatever it may be) of the oscillation remains constant. However, astronomers need to know which modes will grow and which will decay away. This calculation must include the physics involved in Eddington's valve mechanism. The equations describing the transfer of heat and radiation through the stellar layers (similar to those discussed in Section 10.4) must be incorporated in such a *nonadiabatic* computation. These nonadiabatic expressions may also be linearized and solved to obtain the periods and growth rates of the individual modes. However, a more sophisticated and costly *nonlinear*, *nonadiabatic* calculation is needed to reproduce the complicated light and radial velocity

curves that are observed for some variable stars. The computer problem at the end of this chapter asks you to carry out a nonlinear (but still adiabatic) calculation of the pulsation of this one-zone model.

Dynamical Stability

Equation (14.12) provides a very important insight into the **dynamical stability** of a star. If $\gamma < 4/3$, then the right-hand side of Eq. (14.12) is positive. The solution is now $\delta R = Ae^{-\kappa t}$, where κ^2 is the same as ω^2 in Eq. (14.13). Instead of pulsating, the star *collapses* if $\gamma < 4/3$. The increase in gas pressure is not enough to overcome the inward pull of gravity and push the mass shell back out again, resulting in a *dynamically unstable* model. This instability caused by a reduction in the value of γ will be seen again in Section 16.4, where the effect of relativity on white dwarf stars is described.

For the case of *nonadiabatic* oscillations, the time dependence of the pulsation is usually taken to be the real part of $e^{i\sigma t}$, where σ is the complex frequency $\sigma = \omega + i\kappa$. In this expression, ω is the usual pulsation frequency, while κ is a *stability coefficient*. The pulsation amplitude is then proportional to $e^{-\kappa t}$, and $1/\kappa$ is the characteristic time for the growth or decay of the oscillations.

14.4 ■ NONRADIAL STELLAR PULSATION

As some types of stars pulsate, their surfaces do not move uniformly in and out in a simple “breathing” motion. Instead, such a star executes a more complicated type of **nonradial** motion in which some regions of its surface expand while other areas contract.

Nonradial Oscillations and Spherical Harmonic Functions

Figure 14.16 shows the angular patterns for several nonradial modes. If the stellar surface is moving outward within the lighter regions, then it is moving inward within the shaded areas. Scalar quantities such as the change in pressure (δP) follow the same pattern, having positive values in some areas and negative values in others. Formally, these patterns are described by the real parts of the spherical harmonic functions, $Y_\ell^m(\theta, \phi)$, where ℓ is a non-negative integer and m is equal to any of the $2\ell + 1$ integers between $-\ell$ and $+\ell$.⁸ There are ℓ *nodal circles* (where $\delta r = 0$), with $|m|$ of these circles passing through the poles of the star and the remaining $\ell - |m|$ nodal circles being parallel to the star’s equator. If $\ell = m = 0$, then the pulsation is purely radial.

A few examples of $Y_\ell^m(\theta, \phi)$ functions are

$$Y_0^0(\theta, \phi) = K_0^0$$

$$Y_1^0(\theta, \phi) = K_1^0 \cos \theta$$

$$Y_1^{\pm 1}(\theta, \phi) = K_1^{\pm 1} \sin \theta e^{\pm i\phi}$$

⁸Spherical harmonics are often encountered in physics when spherical symmetry is employed. A common example in the undergraduate physics curriculum is the use of spherical harmonics to describe the quantum mechanical wave functions of a hydrogen atom. Recall the discussion in Section 5.4 and the orbitals depicted in Fig. 5.12.

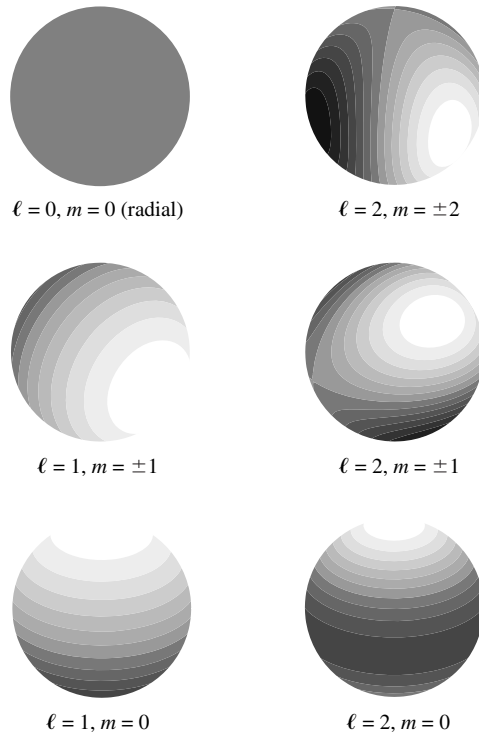


FIGURE 14.16 Nonradial pulsation patterns. The modes of pulsation are represented by the real parts of the spherical harmonic functions, $Y_\ell^m(\theta, \phi)$.

$$Y_2^0(\theta, \phi) = K_2^0(3 \cos^2 \theta - 1)$$

$$Y_2^{\pm 1}(\theta, \phi) = K_2^{\pm 1} \sin \theta \cos \theta e^{\pm i \phi}$$

$$Y_2^{\pm 2}(\theta, \phi) = K_2^{\pm 2} \cos^2 \theta e^{\pm 2i \phi}$$

where the K_ℓ^m s are “normalization” constants and i is the imaginary number $i \equiv \sqrt{-1}$. Recall from Euler’s formula that $e^{\pm mi\phi} = \cos(m\phi) \pm i \sin(m\phi)$. Thus, the real part of $e^{\pm mi\phi}$ is just $\cos(m\phi)$.

The patterns for nonzero m represent *traveling waves* that move across the star parallel to its equator. (Imagine these patterns on a beach ball, with the ball slowly spinning about the vertical axis.) The time required for the waves to travel around the star is $|m|$ times the star’s pulsation period. However, it is important to note that the star itself may not be rotating at all. Just as water waves may travel across the surface of a lake without the water itself making the trip, these traveling waves are disturbances that pass through the stellar gases.⁹

⁹Observations of nonradially pulsating stars are considered in Problem 14.10.

The p and f Modes

In Section 14.2, the radial pulsation of stars was attributed to standing sound waves in the stellar interior. For the case of nonradial oscillations, the sound waves can propagate horizontally as well as radially to produce waves that travel around the star. Because *pressure* provides the restoring force for sound waves, these nonradial oscillations are called **p-modes**. A complete description of a p-mode requires specification of its radial and angular nodes. For example, a p_2 mode may be thought of as the nonradial analog of a radial second overtone mode. The p_2 mode with $\ell = 4$ and $m = -3$ has two radial nodes between the center and the surface, and its angular pattern has four nodal lines, three through the poles and one parallel to the equator. Figure 14.17 shows two p-modes for a $12 M_\odot$ main-sequence star model; you may note the similarities between this figure and Fig. 14.10, with most of the motion occurring near the stellar surface. Also shown is the **f-mode**, which can be thought of as a surface gravity wave (note the rapid rise in amplitude with radius). The frequency of the f-mode is intermediate between the p-modes and the g-modes (discussed later). There is no radial analog for the f-mode.

The Acoustic Frequency

An estimate of the angular frequency of a p-mode may be obtained from the time for a sound wave to travel one horizontal wavelength, from one angular nodal line to the next. This horizontal wavelength is given by the expression

$$\lambda_h = \frac{2\pi r}{\sqrt{\ell(\ell+1)}}, \quad (14.15)$$

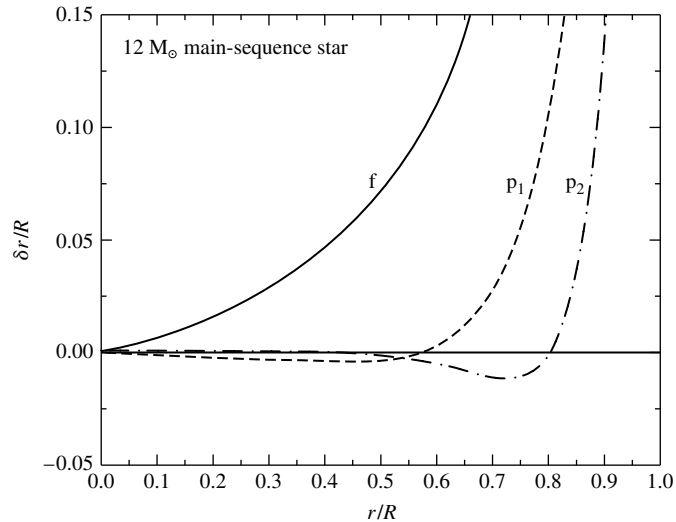


FIGURE 14.17 Nonradial p-modes with $\ell = 2$. The waveforms have been arbitrarily scaled so that $\delta r/R = 1$ at the star's surface. The f-mode is also shown.

where r is the radial distance from the center of the star. The **acoustic frequency** at this depth in the star is then defined as

$$S_\ell = \frac{2\pi}{\text{time for sound to travel } \lambda_h},$$

which can be written as

$$\begin{aligned} S_\ell &= 2\pi \left[\frac{v_s}{2\pi r / \sqrt{\ell(\ell+1)}} \right] \\ &= \sqrt{\frac{\gamma P}{\rho}} \frac{\sqrt{\ell(\ell+1)}}{r}, \end{aligned} \quad (14.16)$$

where v_s is the adiabatic sound speed given by Eq. (10.84). Because the speed of sound is proportional to the square root of the temperature [recall from the ideal gas law, Eq. (10.11), that $P/\rho \propto T$], the acoustic frequency is large in the deep interior of the star and decreases with increasing r . The frequency of a p-mode is determined by the average value of S_ℓ , with the largest contributions to the average coming from the regions of the star where the oscillations are most energetic.

In the absence of rotation, the pulsation period depends only on the number of radial nodes and the integer ℓ . The period is independent of m because with no rotation there are no well-defined poles or equator; thus m has no physical significance. On the other hand, if the star is rotating, the rotation itself defines the poles and equator, and the pulsation frequencies for modes with different values of m become separated or *split* as the traveling waves move either with or against the rotation (the sign of m determines the direction in which the waves move around the star). The amount by which the pulsation frequencies are split depends on the angular rotation frequency, Ω , of the star, with the rotationally produced shift in frequency proportional to the product $m\Omega$ for the simple case of uniform rotation. As we will discuss later, this frequency splitting provides a powerful probe for measuring the rotation of the Sun's interior.

The g Modes

Just as pressure supplies the restoring force for the compression and expansion of the p-mode sound waves, *gravity* is the source of the restoring force for another class of nonradial oscillations called **g-modes**. The g-modes are produced by *internal gravity waves*. These waves involve a “sloshing” back and forth of the stellar gases, which is ultimately connected to the *buoyancy* of stellar material. Because “sloshing” cannot occur for purely radial motion, there are no radial analogs for the g-modes.

The Brunt–Väisälä (Buoyancy) Frequency

To gain a better understanding of this oscillatory motion for g-modes, consider a small bubble of stellar material that is displaced upward from its equilibrium position in the star

by an amount dr , as shown in Fig. 10.10.¹⁰ We will assume that this motion occurs

1. slowly enough that the pressure within the bubble, $P^{(b)}$, is always equal to the pressure of its surroundings, $P^{(s)}$; and
2. rapidly enough that there is no heat exchanged between the bubble and its surroundings.

The second assumption means that the expansion and compression of the gas bubble are *adiabatic*. If the density of the displaced bubble is greater than the density of its new surroundings, the bubble will fall back to its original position. The net restoring force *per unit volume* on the bubble in its final position is the difference between the upward buoyant force (given by Archimedes's law) and the downward gravitational force:

$$f_{\text{net}} = \left(\rho_f^{(s)} - \rho_f^{(b)} \right) g,$$

where $g = GM_r/r^2$ is the local value of the gravitational acceleration. Using a Taylor expansion for the densities about their initial positions results in

$$f_{\text{net}} = \left[\left(\rho_i^{(s)} + \frac{d\rho^{(s)}}{dr} dr \right) - \left(\rho_i^{(b)} + \frac{d\rho^{(b)}}{dr} dr \right) \right] g.$$

The initial densities of the bubble and its surroundings are the same, so these terms cancel, leaving

$$f_{\text{net}} = \left(\frac{d\rho^{(s)}}{dr} - \frac{d\rho^{(b)}}{dr} \right) g dr.$$

Because the motion of the bubble is adiabatic, Eq. (10.87) can be used to replace $d\rho^{(b)}/dr$:

$$f_{\text{net}} = \left(\frac{d\rho^{(s)}}{dr} - \frac{\rho_i^{(b)}}{\gamma P_i^{(b)}} \frac{dP^{(b)}}{dr} \right) g dr.$$

Looking at this equation, all of the “*b*” superscripts may be changed to “*s*” because the initial densities are equal, and according to the first assumption given, the pressures inside and outside the bubble are *always* the same. Thus all quantities in this equation refer to the stellar material surrounding the bubble. With that understanding, the subscripts may be dropped completely, resulting in

$$f_{\text{net}} = \left(\frac{1}{\rho} \frac{d\rho}{dr} - \frac{1}{\gamma P} \frac{dP}{dr} \right) \rho g dr.$$

For convenience, the term in parentheses is defined as

$$A \equiv \frac{1}{\rho} \frac{d\rho}{dr} - \frac{1}{\gamma P} \frac{dP}{dr}. \quad (14.17)$$

¹⁰The following discussion is just a reexamination of the problem of convection, last seen in Section 10.4, from another perspective.

Thus the net force per unit volume acting on the bubble is

$$f_{\text{net}} = \rho A g dr. \quad (14.18)$$

If $A > 0$, the net force on the displaced bubble has the same sign as dr , and so the bubble will continue to move away from its equilibrium position. This is the condition necessary for *convection* to occur, and it is equivalent to the other requirements previously found for convective instability, such as Eq. (10.94). However, if $A < 0$, then the net force on the bubble will be in a direction opposite to the displacement, and so the bubble will be pushed back toward its equilibrium position. In this case, Eq. (14.18) has the form of Hooke's law, with the restoring force proportional to the displacement. Thus if $A < 0$, the bubble will oscillate about its equilibrium position with simple harmonic motion.

Dividing the force per unit volume, f_{net} , by the mass per unit volume, ρ , gives the force per unit mass, or acceleration: $a = f_{\text{net}}/\rho = A g dr$. Because the acceleration is simply related to the displacement for simple harmonic motion,¹¹ we have

$$a = -N^2 dr = A g dr,$$

where N is the angular frequency of the bubble about its equilibrium position, called the **Brunt–Väisälä frequency** or the **buoyancy frequency**,

$$N = \sqrt{-Ag} = \sqrt{\left(\frac{1}{\gamma P} \frac{dP}{dr} - \frac{1}{\rho} \frac{d\rho}{dr} \right) g}. \quad (14.19)$$

The buoyancy frequency is zero at the center of the star (where $g = 0$) and at the edges of convection zones (where $A = 0$). Recall that $A < 0$ where there is no convection, so N is larger in regions that are more stable against convection. Inside a convection zone, where $A > 0$, the buoyancy frequency is not defined.

The g and p Modes as Probes of Stellar Structure

The “sloshing” effect of neighboring regions of the star produces the internal gravity waves that are responsible for the g-modes of a nonradially pulsating star. The frequency of a g-mode is determined by the value of N averaged across the star. Figure 14.18 shows several g-modes for the same stellar model that was used for Fig. 14.17. A comparison of these two figures reveals significant differences between these classes of modes, making them very useful to astronomers attempting to study the interior of the Sun and other stars. Most important, notice the difference in the vertical scales of the two figures. The g-modes involve significant movement of the stellar material deep within the star, while the p-mode's motions are confined near the stellar surface. Thus g-modes provide a view into the very heart of a star, while p-modes allow a diagnosis of the conditions in its surface layers.

¹¹For example, recall that $F = ma = -kx$ for a spring. The acceleration is $a = -\omega^2 x$, where $\omega = \sqrt{k/m}$ is the angular frequency of the spring's motion.

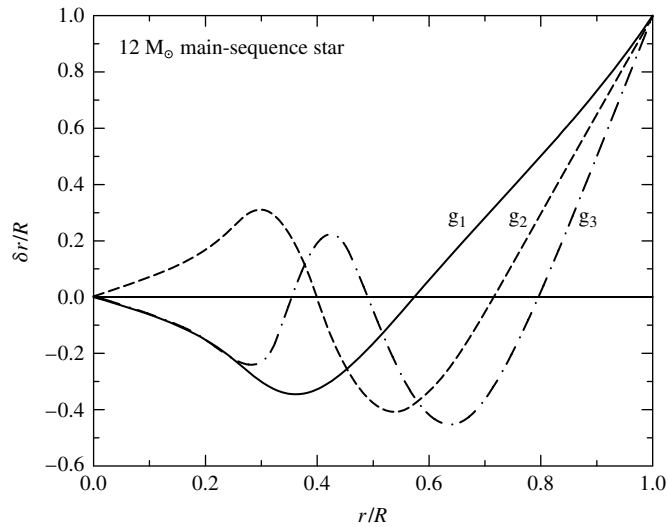


FIGURE 14.18 Nonradial g-modes with $\ell = 2$. The waveforms have been arbitrarily scaled so that $\delta r/R = 1$ at the star's surface.

14.5 ■ HELIOSEISMOLOGY AND ASTEROSEISMOLOGY

All of the ideas of nonradial pulsation come into play in the science of **helioseismology**, the study of the oscillations of the Sun first observed in 1962 by American astronomers Robert Leighton (1919–1997), Robert Noyes, and George Simon. A typical solar oscillation mode has a very low amplitude, with a surface velocity of only 0.10 m s^{-1} or less,¹² and a luminosity variation $\delta L/L_{\odot}$ of only 10^{-6} . With an incoherent superposition of roughly *ten million* modes rippling through its surface and interior, our star is “ringing” like a bell.

The Five-Minute Solar Oscillations

The oscillations observed on the Sun have modes with periods between three and eight minutes and very short horizontal wavelengths (ℓ ranging from 0 to 1000 or more). These so-called *five-minute oscillations* have been identified as p-modes. The five-minute p-modes are concentrated below the photosphere within the Sun's convection zone; Fig. 14.19 shows a typical p-mode. g-modes are located deep in the solar interior, below the convection zone. By studying these p-mode oscillations, astronomers have been able to gain new insights into the structure of the Sun in these regions.¹³

¹²These incredibly precise velocity measurements are made by carefully observing the Doppler shifts of spectral absorption lines such as Fe I (557.6099 nm) through a narrow slit that follows the rotating solar surface.

¹³A series of 160-minute “g-modes” were believed to have been observed as well. However, continuous observations over 690 days using the GOLF instrument onboard the SOHO spacecraft were unable to detect any evidence of the controversial mode. It is believed that ground-based observations that indicated a 160-minute mode were due to harmonic effects associated with Earth's atmosphere; note that 160 minutes is exactly 1/9 of the 24-hour solar day.

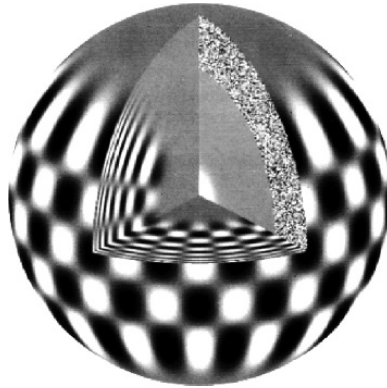


FIGURE 14.19 Five-minute p_{15} mode with $\ell = 20$ and $m = 16$. The solar convection zone is the stippled region, where the p-modes are found. (Courtesy of National Optical Astronomy Observatories.)

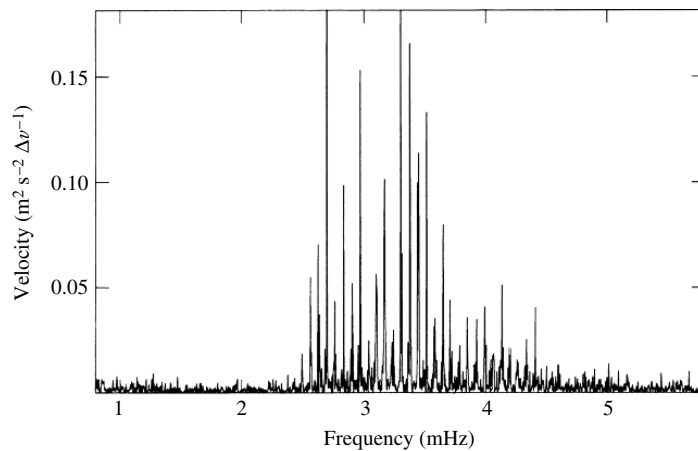


FIGURE 14.20 Relative power of solar p-modes; a period of five minutes corresponds to a frequency of 3.33 mHz. (Figure adapted from Grec, Fossat, and Pomerantz, *Nature*, 288, 541, 1980.)

Figure 14.20 shows the relative power contained in the solar p-modes. This information can also be plotted in another manner, as shown in Fig. 14.21, with ℓ on the horizontal axis and the pulsation frequency on the vertical axis. Circles show the observed frequencies, and each continuous ridge corresponds to a different p-mode (p_1 , p_2 , p_3 , etc.). The superimposed lines are the *theoretical* frequencies calculated for a solar model. All of the observed five-minute modes have been identified in this way. The fit is certainly impressive but not quite exact. A solar model must be carefully tuned to obtain the best agreement between the theoretical and observed p-mode frequencies. This procedure can reveal much about the depth of the solar convection zone and about the rotation and composition of the outer layers of the Sun.

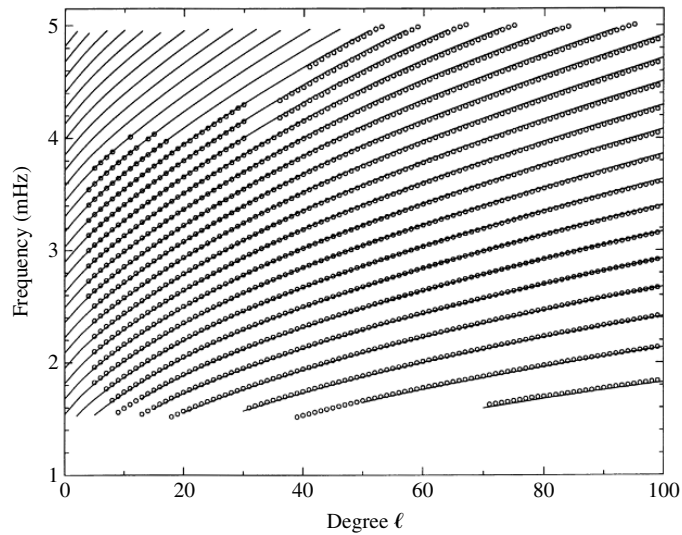


FIGURE 14.21 Solar p-modes: observations (circles) and theory (lines). (Figure adapted from Libbrecht, *Space Sci. Rev.*, 47, 275, 1988.)

Differential Rotation and the Solar Convection Zone

As mentioned in Chapter 11, based on helioseismology studies, combined with detailed stellar evolution calculations, the base of the solar convection zone is known to be located at $0.714 R_{\odot}$, with a temperature of about 2.18×10^6 K. The rotational splitting observed for p-mode frequencies indicates that differential rotation observed at the Sun's surface decreases slightly down through the convection zone (recall Fig. 11.16). Those p-modes with shorter horizontal wavelengths (larger ℓ) penetrate less deeply into the convection zone, so the difference in rotational frequency splitting with ℓ reveals the depth dependence of the rotation. The measurement of the variation in rotation with the distance from the solar equator comes from the dependence of the rotational frequency splitting on m . Below the convection zone, the equatorial and polar rotation rates converge to a single value at $r/R_{\odot} \approx 0.65$. Because a change in the rotation rate with depth is needed to convert the Sun's magnetic field from a poloidal to a toroidal geometry (as discussed in Section 11.3), these results indicate that the Sun's magnetic dynamo is probably seated in the tachocline at the interface between the radiation zone and the convection zone.

Tests of Composition

The abundance of helium in the outer layers of the Sun can also be inferred from a comparison of the observed and theoretical p-mode ridges in Fig. 14.21. The results are consistent with a value of $Y = 0.2437$ for the mass fraction of helium at the Sun's surface.

Probing the Deep Interior

Astronomers have experienced more difficulty in their attempts to use the solar g-modes as a probe of the Sun's interior. Because the g-modes dwell beneath the convection zone,

their amplitudes are significantly diminished at the Sun's surface. To date, no definite identification of g-modes has been made. Nevertheless, the potential rewards of using these oscillations to learn more about the core of the Sun compel astronomers to apply their observational ingenuity to these g-modes.

Driving Solar Oscillations

The question of the mechanism responsible for driving the solar oscillations has not yet been conclusively answered. Our main-sequence Sun is not a normal pulsating star. It lies far beyond the red edge of the instability strip on the H–R diagram (see Fig. 14.8) where turbulent convection overcomes the tendency of the ionization zones to absorb heat at maximum compression. Eddington's valve mechanism thus cannot be responsible for the solar oscillations. However, the timescale for convection near the top of the convection zone is a few minutes, and it is strongly suspected that the p-modes are driven by tapping into the turbulent energy of the convection zone itself, where the p-modes are confined.

δ Scuti Stars and Rapidly Oscillating Ap Stars

The techniques of helioseismology can be applied to other stars as well. **Asteroseismology** is the study of the pulsation modes of stars in order to investigate their internal structures, chemical composition, rotation, and magnetic fields.

δ Scuti stars are Population I main-sequence stars and giant stars in the spectral class range A to F. They tend to pulsate in low-overtone radial modes, as well as in low-order p-modes (and possibly g-modes). The amplitudes of δ Scutis are fairly small, ranging from a few mmag to roughly 0.8 mag. Population II subgiants also exhibit radial and nonradial oscillations and are known as SX Phoenicis stars.

Another interesting class of pulsating stars are the **rapidly oscillating Ap stars** (roAp), found in the same portion of the H–R diagram as the δ Scuti stars. These stars have peculiar surface chemical compositions (hence the “p” designation), are rotating, and have strong magnetic fields. The unusual chemical composition is likely due to settling of heavier elements, similar to the elemental diffusion that has occurred near the surface of the Sun. Some elements may also have been elevated in the atmosphere if they have a significant number of absorption lines near the peak of the star's blackbody spectrum. These atoms preferentially absorb photons that impart a net upward momentum. If the atmosphere is sufficiently stable against turbulent motions, some of these atoms will tend to drift upward.

roAp stars have very small pulsation amplitudes of less than 0.016 mag. It appears that they primarily pulsate in higher-order p-modes and that the axis for the pulsation is aligned with the magnetic field axis, which is tilted somewhat to the rotation axis (an oblique rotator model). roAp stars are among the most well-studied of main-sequence stars other than the Sun, but the pulsation driving mechanism still remains in question.

SUGGESTED READING

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PROBLEMS

- 14.1** Use the light curve for Mira, Fig. 14.1, to estimate the ratio of Mira’s luminosity at visible wavelengths, when it is brightest to when it is dimmest. For what fraction of its pulsation cycle is Mira visible to the naked eye?
- 14.2** If the intrinsic uncertainty in the period–luminosity relation shown in Fig. 14.5 is $\Delta M \approx 0.5$ magnitude, find the resulting fractional uncertainty in the calculated distance to a classical Cepheid.
- 14.3** Several remote classical Cepheids were discovered in 1994 by the Hubble Space Telescope in the galaxy denoted M100. (M100 is a member of the Virgo cluster, a rich cluster of galaxies.) Figure 14.22 shows the period–luminosity relation for these Cepheids. Use the two Cepheids nearest the figure’s best-fit line to estimate the distance to M100. The mean visual extinction is $A_V = 0.15 \pm 0.17$ magnitudes for the M100 Cepheids. Compare your result to the distance of 17.1 ± 1.8 Mpc obtained by Wendy Freedman and her colleagues. You are referred to Freedman et al. (1994) for more information on the discovery and importance of these remote pulsating stars.
- 14.4** Make a graph similar to Fig. 14.5 showing the period–luminosity relation for both the classical Cepheids and W Virginis stars.

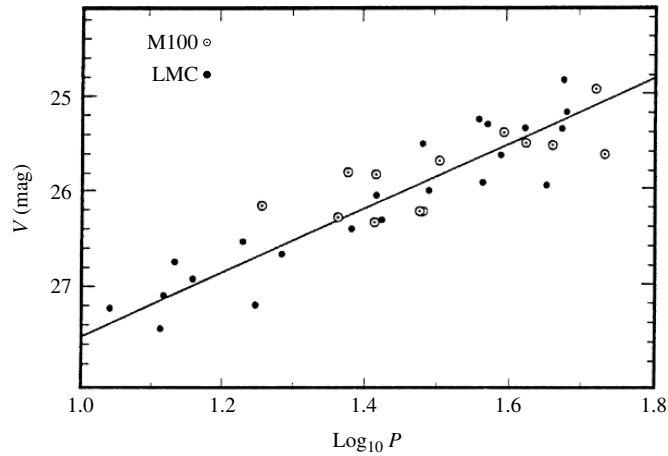


FIGURE 14.22 A composite period–luminosity relation for Problem 14.3. The white circles denote Cepheids in M100, and the black circles show nearby Cepheids found in the Large Magellanic Cloud (a small galaxy that neighbors our Milky Way Galaxy). The average visual magnitudes of the LMC Cepheids have been increased by the same amount to match those of the M100 variables. The required increase in V for a best fit is then used to find the relative distances to the LMC and M100. (Adapted from Freedman et al., *Nature*, 371, 757, 1994.)

- 14.5** Assuming (incorrectly) that the oscillations of δ Cephei are sinusoidal, calculate the greatest excursion of its surface from its equilibrium position.
- 14.6** Use Eq. (14.6) to estimate the pulsation period that the Sun would have if it were to oscillate radially.
- 14.7** Derive Eq. (14.11) by linearizing the adiabatic relation

$$P V^\gamma = \text{constant}.$$

- 14.8 (a)** Linearize the Stefan–Boltzmann equation in the form of Eq. (3.17) to show that

$$\frac{\delta L}{L_0} = 2 \frac{\delta R}{R_0} + 4 \frac{\delta T}{T_0}.$$

- (b)** Linearize the adiabatic relation $T V^{\gamma-1} = \text{constant}$, and so find a relation between $\delta L/L_0$ and $\delta R/R_0$ for a spherical blackbody model star composed of an ideal monatomic gas.
- 14.9** Consider a general potential energy function, $U(r)$, for a force $\mathbf{F} = -(dU/dr)\hat{\mathbf{r}}$ on a particle of mass m . Assume that the origin ($r = 0$) is a point of stable equilibrium. By expanding $U(r)$ in a Taylor series about the origin, show that if a particle is displaced slightly from the origin and then released, it will undergo simple harmonic motion about the origin. This explains why the linearization procedure of Section 14.3 is guaranteed to result in sinusoidal oscillations.
- 14.10** Figure 14.23 shows a view of a hypothetical nonradially pulsating ($\ell = 2, m = -2$), rotating star from above the star's north pole. From the vantage point of Earth, astronomers view the star along its equatorial plane. Assuming that a spectral absorption line appears as in Fig. 9.18 when the bottom of Fig. 14.23 is facing Earth, sketch the changes in the appearance of the

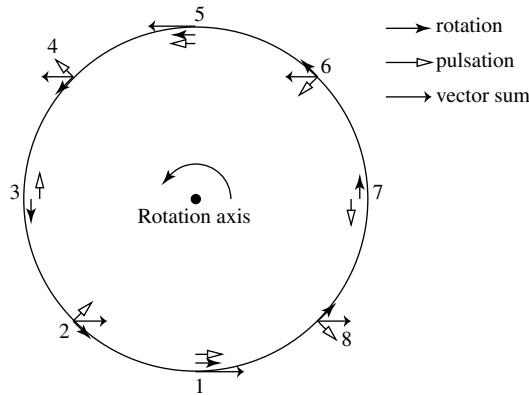


FIGURE 14.23 Surface velocities for a rotating, pulsating star ($\ell = 2$, $m = -2$) for Problem 14.10. The arrows indicate the surface velocities due to rotation alone, pulsation alone, and their vector sum.

line profile due to Doppler shifts caused by the total surface velocity as the star rotates. (Don't worry about the timing; just sketch the spectral line as seen from the eight different points of view shown that are directly over the star's equator.) Assume that the equivalent width of the line does not change. You may wish to compare your line profiles with those actually observed for a nonradially pulsating star such as the β Cephei star 12 Lacertae; see Smith, *Ap. J.*, 240, 149, 1980. For convenience, the magnitudes of the rotation and pulsation velocities are assumed to be equal.

- 14.11** Show that Eq. (10.94), the condition for convection to occur, is the same as the requirement that $A > 0$, where A is given by Eq. (14.17). Assume that the mean molecular weight, μ , does not vary.
- 14.12** In a convection zone, the timescale for convection (see Section 10.4) is related to the value of A (Eq. 14.17) by

$$t_c \simeq 2\sqrt{2/Ag}.$$

Table 14.2 shows the values of the pressure and density at two points near the top of the Sun's convection zone as described by a solar model. Use these values and $\gamma = 5/3$ to obtain an estimate of the timescale for convection near the top of the Sun's convection zone. How does your answer compare with the range of periods observed for the Sun's p-modes?

TABLE 14.2 Data from a Solar Model for Problem 14.12. (Data from Joyce Guzik, private communication.)

r (m)	P (N m ⁻²)	ρ (kg m ⁻³)
6.959318×10^8	9286.0	2.2291×10^{-4}
6.959366×10^8	8995.7	2.1925×10^{-4}

COMPUTER PROBLEM

14.13 In this problem you will carry out a nonlinear calculation of the radial pulsation of the one-zone model described in Example 14.3.1. The equations that describe the oscillation of this model star are Newton's second law for the forces on the shell,

$$m \frac{dv}{dt} = -\frac{GMm}{R^2} + 4\pi R^2 P, \quad (14.20)$$

and the definition of the velocity, v , of the mass shell,

$$v = \frac{dR}{dt}. \quad (14.21)$$

As in Example 14.3.1, we assume that the expansion and contraction of the gas are adiabatic:

$$P_i V_i^\gamma = P_f V_f^\gamma, \quad (14.22)$$

where the “initial” and “final” subscripts refer to any two instants during the pulsation cycle.

- (a) Explain in words the meaning of each term in Eq. (14.20).
 (b) Use Eq. (14.22) to show that

$$P_i R_i^{3\gamma} = P_f R_f^{3\gamma}. \quad (14.23)$$

- (c) You will not be taking derivatives. Instead, you will take the difference between the initial and final values of the radius R and radial velocity v of the shell divided by the time interval Δt separating the initial and final values. That is, you will use $(v_f - v_i)/\Delta t$ instead of dv/dt , and $(R_f - R_i)/\Delta t$ instead of dR/dt in Eqs. (14.20) and (14.21). A careful analysis shows that you should use $R = R_i$ and $P = P_i$ on the right-hand side of Eq. (14.20), and use $v = v_f$ on the left-hand side of Eq. (14.21). Make these substitutions in Eqs. (14.20) and (14.21), and show that you can write

$$v_f = v_i + \left(\frac{4\pi R_i^2 P_i}{m} - \frac{GM}{R_i^2} \right) \Delta t \quad (14.24)$$

and

$$R_f = R_i + v_f \Delta t. \quad (14.25)$$

- (d) Now you are ready to calculate the oscillation of the model star. The mass of a typical classical Cepheid is $M = 1 \times 10^{31}$ kg (5 $M_\odot$), and the mass of the surface layers may be arbitrarily assigned $m = 1 \times 10^{26}$ kg. For starting values at time $t = 0$, take

$$R_i = 1.7 \times 10^{10} \text{ m}$$

$$v_i = 0 \text{ m s}^{-1}$$

$$P_i = 5.6 \times 10^4 \text{ N m}^{-2}$$

and use a time interval of $\Delta t = 10^4$ s. Take the ratio of specific heats to be $\gamma = 5/3$ for an ideal monatomic gas. Use Eq. (14.24) to calculate the final velocity v_f at the end of one time interval (at time $t = 1 \times 10^4$ s); then use Eq. (14.25) to calculate the final radius R_f and Eq. (14.23) to calculate the final pressure P_f . Now take these final values to be your

new initial values, and find new values for R , v , and P after two time intervals (at time $t = 2 \times 10^4$ s). Continue to find R , v , and P for 150 time intervals, until $t = 1.5 \times 10^6$ s. Make three graphs of your results: R vs. t , v vs. t , and P vs. t . Plot the time on the horizontal axis.

- (e) From your graphs, measure the period Π of the oscillation (both in seconds and in days) and the equilibrium radius, R_0 , of the model star. Compare this value of the period with that obtained from Eq. (14.14). Also compare your results with the period and radial velocity observed for δ Cephei.

The Fate of Massive Stars

- 15.1 *Post-Main-Sequence Evolution of Massive Stars*
- 15.2 *The Classification of Supernovae*
- 15.3 *Core-Collapse Supernovae*
- 15.4 *Gamma-Ray Bursts*
- 15.5 *Cosmic Rays*

15.1 ■ POST-MAIN-SEQUENCE EVOLUTION OF MASSIVE STARS

Astronomers have been observing the fascinating Southern Hemisphere star η Carinae ($\alpha = 10^{\text{h}} 45^{\text{m}} 03.59^{\text{s}}$, $\delta = -59^{\circ} 41' 04.26''$) since at least 1600. Between 1600 and about 1830, observers had reported the star as being about second magnitude, although it was sometimes reported to be a fourth-magnitude star. Then, in 1820 or 1830, it may have started becoming more active. However, in 1837 η Car suddenly brightened significantly, fluctuating between zero and first magnitude for about twenty years. At one point, its magnitude reached about -1 , making it the second brightest extra-Solar-System object in the sky (only Sirius was brighter). During this period John Herschel (1792–1871) described η Carinae as being “fitfully variable.”

η Car’s remarkable brightness achieved in 1837 is all the more impressive given that the star is approximately 2300 pc from Earth (by comparison, Sirius is only 2.64 pc from Earth). After 1856, this mysterious star began to fade again, dropping to about eighth magnitude by 1870. With the exception of a lesser brightening event that took place between 1887 and 1895, η Car has been relatively quiet since the “Great Eruption” between 1837 and 1856. Over the past century and a half, η Car has brightened slightly, and it is currently at a visual magnitude of about six.

Another star in the Milky Way Galaxy has behaved similarly. P Cygni was apparently too faint to be seen by the naked eye prior to 1600 but then suddenly appeared, reaching third magnitude. Following this early eruption, P Cyg faded from view, only to reappear in 1655, becoming nearly as bright as it was in 1600. P Cyg has been a roughly constant fifth-magnitude star since 1700, although it may have brightened slightly over the past several centuries. [You may recall that spectral line profiles known as P Cygni profiles (see Fig. 12.17), named for this episodic star, are indicative of mass loss.]

Luminous Blue Variables

A small number of other stars are known to behave similarly to η Carinae and P Cygni, some in our galaxy and some farther away. S Doradus, located in the Large Magellanic Cloud (LMC), a satellite galaxy of the Milky Way, is perhaps the best-known extragalactic example. Similar stars were discovered in nearby galaxies by Edwin Hubble and Allan Sandage. This class of stars is referred to by several different names, including **S Doradus variables**, **Hubble–Sandage variables**, and **luminous blue variables** (LBVs). In this text, we will adopt the designation of LBV for this class of stars.

While perhaps an extreme example of an LBV, η Car is certainly the best-studied representative of the class. A Hubble Space Telescope image of η Car is shown in Fig. 15.1. Its bipolar structure, known as the “homunculus,” is clearly evident, as is its equatorial disk. From Doppler measurements, the lobes are expanding outward at about 650 km s^{-1} , although a number of different velocities can be recorded along any particular line of sight. The expanding lobes are largely hollow, but the material in the shells contains molecules of H_2 , CH, and OH. However, it appears that the homunculus is significantly depleted in C and O, while being enriched in He and N. This would suggest that the ejected material has undergone nuclear processing by the CNO cycle (see Eqs. 10.48–10.57). The present rate of mass loss from η Car is on the order of $10^{-3} M_{\odot} \text{ yr}^{-1}$, but it probably ejected one to three solar masses of material during the twenty years of the Great Eruption.

During the Great Eruption, η Car’s luminosity may have been about $2 \times 10^7 L_{\odot}$, whereas its present quiescent luminosity is near $5 \times 10^6 L_{\odot}$. It is also estimated that the central star’s effective temperature is roughly 30,000 K. Although at the time this text was written, $m_V \sim 6$ for η Car, most of its luminosity is initially emitted in the ultraviolet wavelength region, owing to the high effective temperature. Much of the UV radiation is scattered, absorbed, and re-emitted by dust grains in the infrared portion of the electromagnetic

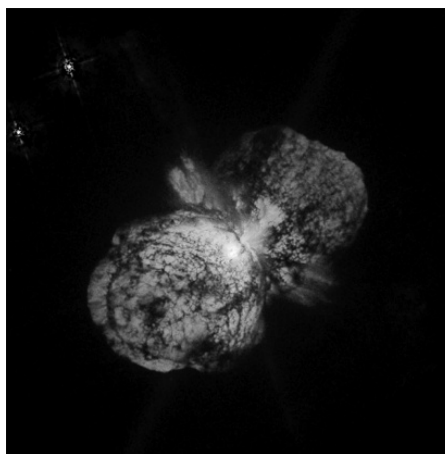


FIGURE 15.1 η Carinae is a luminous blue variable that is estimated to have a mass of $120 M_{\odot}$ and is rapidly losing mass. Each lobe has a diameter of approximately 0.1 pc. [Courtesy of Jon Morse (University of Colorado) and NASA.]

spectrum at temperatures ranging from 200 K to 1000 K. The visual magnitude alone leads to an underestimate of the total luminosity of the object today.

As a class, LBVs tend to have high effective temperatures of between 15,000 K and 30,000 K, with luminosities in excess of $10^6 L_{\odot}$. This places LBVs in the upper-left-hand portion of the H–R diagram. Given the composition of their atmospheres and ejecta, LBVs are clearly evolved, post-main-sequence stars. It also appears that LBVs cluster in an instability region of the H–R diagram, suggesting that their behavior is transient, turning on shortly after they leave the main sequence and ceasing after some period of time.

A variety of mechanisms have been proposed to explain the behavior of LBVs, including their variability and dramatic mass loss. As discussed in Section 10.6, the upper end of the main sequence is very near the Eddington luminosity limit where the force due to radiation pressure may equal or exceed the force of gravity on surface layers of the star. The derived expression for the Eddington luminosity limit, Eq. (10.114), is a function of the Rosseland mean opacity of the surface layers,

$$L_{\text{Ed}} = \frac{4\pi Gc}{\bar{\kappa}} M.$$

The “classical” Eddington limit assumes that the opacity is due entirely to scattering from free electrons (Eq. 9.27), which is constant for a completely ionized gas. A “modified” Eddington limit has been proposed by Roberta M. Humphreys and Kris Davidson, in which some temperature-dependent component of opacity, perhaps due to iron lines, modifies the opacity term as the star evolves to the right in the H–R diagram. As the temperature decreases and the opacity increases, the Eddington luminosity would drop below the actual luminosity of the star, implying that radiation pressure dominates gravity, driving mass loss from the envelope.

A second suggestion is that atmospheric pulsation instabilities may develop, much like those in Cepheids, RR Lyraes, and long-period variables. Some preliminary nonlinear pulsation studies have suggested that large-amplitude oscillations can develop in LBVs, which could conceivably drive mass loss as outwardly moving mass shells are lifted off the surface during the pulsation cycle. Moreover, such pulsations are likely to be very irregular in such a weakly bound atmosphere. Unfortunately, these models are very sensitive to the treatment of time-dependent convection, which is poorly understood in the context of stellar pulsations.

Also intriguing is the apparent high rotation velocity of at least some LBVs. Rapid rotation would result in decreasing the “effective” gravity at the equator of these stars due to centrifugal effects, making the gases in the atmospheres of the equatorial regions easier to drive away from the surface. It has been suggested that the equatorial disk around η Car could have formed from just such an effect during the lesser eruption between 1887 and 1895.

The possibility that LBVs are members of binary star systems has also been suggested as influencing the behavior of these stars. Interestingly, η Car exhibits a 5.54-yr periodicity in the equivalent widths of some of its spectral lines, hinting at the presence of a binary companion, although it is unclear how the presence of a companion can cause the effects observed.

It may turn out that more than one of the mechanisms discussed above could influence the behavior of LBVs, or perhaps the principal mechanism is one that has not yet been identified.

Wolf-Rayet Stars

Closely related to the LBVs are the **Wolf-Rayet stars** (WR). The first WRs were discovered by C. J. E. Wolf and G. Rayet while working at the Paris Observatory in 1867. Using a visual-wavelength spectrometer to conduct a survey of stars in Cygnus, they observed three stars all within one degree of each other that exhibited unusually strong, very broad emission lines, rather than the absorption lines usually seen in other stars. Today, more than 220 WR stars have been identified in the Milky Way Galaxy, although the total number of WRs in the Galaxy is estimated to be between 1000 and 2000 on the basis of sampling statistics (see Fig. 15.2).

Along with the strong emission lines, WR stars are very hot, with effective temperatures of 25,000 K to 100,000 K. WRs are also losing mass at rates in excess of $10^{-5} M_{\odot} \text{ yr}^{-1}$ with wind speeds ranging from 800 km s^{-1} to more than 3000 km s^{-1} . In addition, there is strong evidence that many, and perhaps all, WR stars are rapidly rotating, with equatorial rotation speeds of typically 300 km s^{-1} .

Whereas LBVs are all very massive stars of $85 M_{\odot}$ or more, WRs can have progenitor masses as low as $20 M_{\odot}$. WRs also do not demonstrate the dramatic variability that is characteristic of LBVs.

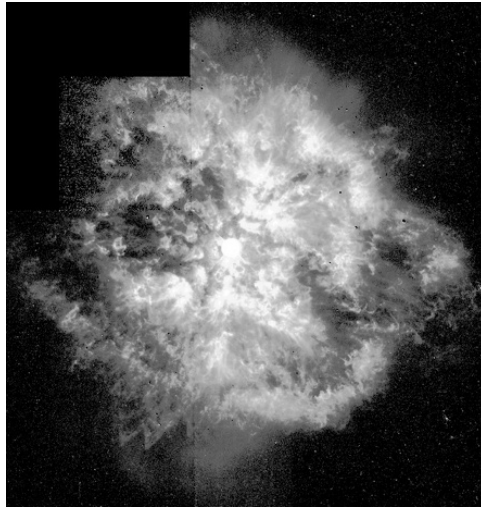


FIGURE 15.2 The nebula M1-67 around the Wolf-Rayet star WR 124. The surface temperature of the star is about 50,000 K. Clumpiness is clearly evident in the nebula, and the mass of each blob is about $30 M_{\oplus}$. WR 124 is at a distance of 4600 pc in Sagittarius. [Courtesy of Yves Grosdidier (University of Montreal and Observatoire de Strasbourg), Anthony Moffat (Université de Montréal), Gilles Joncas (Université Laval), Agnes Acker (Observatoire de Strasbourg), and NASA.]

What really sets Wolf–Rayet stars apart from other stars is their unusual spectra. Not only are the spectra dominated by broad emission lines, but they also reveal a composition that is decidedly atypical. Today we recognize three classes of WR stars: WN, WC, and WO. The spectra of WNs are dominated by emission lines of helium and nitrogen, although emission from carbon, oxygen, and hydrogen is detectable in some WN stars. WC stars exhibit emission lines of helium and carbon, with a distinct absence of nitrogen and hydrogen lines. Finally, the WO stars, which are much rarer than either WNs or WCs, have spectra containing prominent oxygen lines, with some contribution from highly ionized species.

The literature further sub-classifies WN and WC stars based on the degree of ionization of species in the atmosphere. For example, WN2 stars show spectral lines of He II, N IV, and O VI, and WN9 stars contain spectra of low-ionization species such as He I and N III. “Early” (E) and “late” (L) types are also mentioned; WNE stars are Wolf–Rayet stars of ionization classes, WN2 to WN5, and WNL stars are of ionization classes WN6 through WN11. Similarly WC4 stars have higher ionization levels (He II, O IV, C VI), and WC9 stars exhibit lower ionization levels (e.g., He I and C II). WCEs range for WC4 through WC6, and WCLs include WC7 through WC9.

This strange trend in composition from WN to WC to WO was eventually recognized to be a direct consequence of the mass loss of these stars. WNs have lost virtually all of their hydrogen-dominated envelopes, revealing material synthesized by nuclear reactions in the core. Convection in the core of the star has brought equilibrium CNO cycle-processed material to the surface. Further mass loss results in the ejection of the CNO processed material, exposing helium-burning material generated by the triple alpha process (Eqs. 10.60 and 10.61). Then, if the star survives long enough, mass loss will eventually strip away all but the oxygen component of the triple-alpha ash.

In addition to LBVs and WRs, the upper portion of the H–R diagram also contains **blue supergiant stars** (BSG), **red supergiant stars** (RSG), and **Of stars** (O supergiants with pronounced emission lines).

A General Evolutionary Scheme for Massive Stars

In a scheme originally suggested by Peter Conti in 1976 and subsequently modified, a general evolutionary path for massive stars has been outlined. In each case the star ends its life in a **supernova** (SN) explosion, to be discussed in detail in Section 15.3. (The masses listed below are only approximate.)¹

$$\begin{aligned}
 M > 85 M_{\odot} : & \text{O} \rightarrow \text{Of} \rightarrow \text{LBV} \rightarrow \text{WN} \rightarrow \text{WC} \rightarrow \text{SN} \\
 40 M_{\odot} < M < 85 M_{\odot} : & \text{O} \rightarrow \text{Of} \rightarrow \text{WN} \rightarrow \text{WC} \rightarrow \text{SN} \\
 25 M_{\odot} < M < 40 M_{\odot} : & \text{O} \rightarrow \text{RSG} \rightarrow \text{WN} \rightarrow \text{WC} \rightarrow \text{SN} \\
 20 M_{\odot} < M < 25 M_{\odot} : & \text{O} \rightarrow \text{RSG} \rightarrow \text{WN} \rightarrow \text{SN} \\
 10 M_{\odot} < M < 20 M_{\odot} : & \text{O} \rightarrow \text{RSG} \rightarrow \text{BSG} \rightarrow \text{SN}
 \end{aligned}$$

This qualitative evolutionary scheme has been supported by detailed numerical evolutionary models of massive star formation. Evolutionary tracks for stars of solar composition

¹This version has been adopted from Massey, *Annu. Rev. Astron. Astrophys.*, 41, 15, 2003.

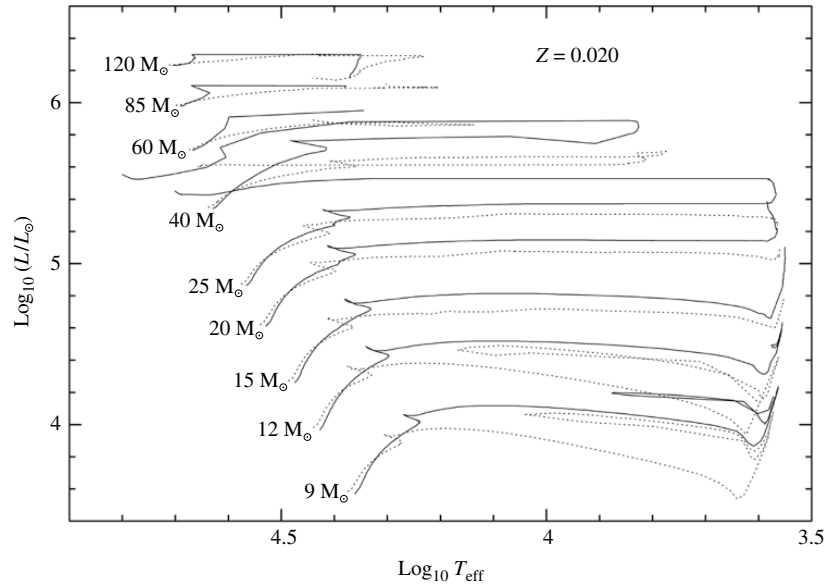


FIGURE 15.3 The evolution of massive stars with $Z = 0.02$. The solid lines are evolutionary tracks computed with initial rotation velocities of 300 km s^{-1} , and the dotted lines are evolutionary tracks for stars without rotation. Mass loss has been included in the models and significantly impacts the evolution of these stars. (Figure from Meynet and Maeder, *Astron. Astrophys.*, 404, 975, 2003.)

ranging in mass from $9 M_{\odot}$ to $120 M_{\odot}$ are shown in Fig. 15.3. These models of Georges Meynet and Andrés Maeder include mass loss typical of massive stars. The models are also computed with and without rotation; when rotation is included, the equatorial rotation speed is taken to be 300 km s^{-1} . Meynet and Maeder point out that rotation can have an appreciable affect on stellar evolution, including driving internal mixing and enhancing mass loss.

The Humphreys–Davidson Luminosity Limit

These massive-star evolutionary tracks indicate that the most massive stars never evolve to the red supergiant portion of the H–R diagram. This is in agreement with the qualitative evolutionary scenario presented above, and it is also consistent with observations. Humphreys and Davidson were the first to point out that there is an upper-luminosity cut-off in the H–R diagram that includes a diagonal component running from highest luminosities and effective temperatures to lower values in both parameters. At that point, when full redward evolutionary tracks develop for stars below about $40 M_{\odot}$, the **Humphreys–Davidson luminosity limit** continues at constant luminosity.

Although very massive stars are extremely rare (only one $100 M_{\odot}$ star exists for every one million $1 M_{\odot}$ stars), they play a major role in the dynamics and chemical evolution of the interstellar medium. The tremendous amount of kinetic energy deposited in the ISM through the stellar winds of massive stars has a significant impact on the kinematics of the ISM. In fact, when very massive stars form, they have the ability to quench star formation

in their regions. The ultraviolet light from massive stars also ionizes gas clouds in their region. And, in addition, the highly enriched gases of massive stellar winds increase the metal content of the ISM, resulting in the formation of increasingly metal-rich stars. Besides being spectacular and exotic objects, massive stars are critically important to the evolution of the galaxies in which they reside.

15.2 ■ THE CLASSIFICATION OF SUPERNOVAE

In A.D. 1006 an extremely bright star suddenly appeared in Lupus. Reaching an estimated apparent visual magnitude of $m_V = -9$, it was reportedly bright enough to read by at night. This event was recorded by astrologers in Europe, China, Japan, Egypt, and Iraq. Based on their writings, it is likely that **Supernova 1006** (SN 1006) appeared about April 30, 1006, and faded from view roughly one year later.

Other, similar events have been seen throughout human history, although only rarely. Perhaps the most famous celestial event of its kind occurred on July 4, 1054, only 48 years after the A.D. 1006 event, when a “guest star” appeared in the night sky in the constellation of Taurus. Yang Wei-T’u, a court astrologer during China’s Sung dynasty, recorded the remarkable event, noting that “after more than a year it gradually became invisible.” In addition to having been carefully documented in the official records of the Sung dynasty, the star was noted by the Japanese and Koreans and was also recorded in an Arabic medical textbook. There is evidence, although this is the subject of some debate, that Europeans may have witnessed the event as well. As with the A.D. 1006 event, this amazing star was visible during daylight. With the development of powerful telescopes, modern astronomers have identified a rapidly expanding cloud, known as the **Crab supernova remnant** at the reported location of this ancient “guest star” (see Fig. 15.4).

It was five hundred years before another star suddenly appeared in the heavens in such dramatic fashion. Tycho Brahe [the most famous astronomer of his day; recall Fig. 2.1(a)] and others witnessed a supernova in A.D. 1572. This strange occurrence was clearly in contrast to the widely held belief in the Western world at the time that the heavens were unchanging. Not to be outdone, his student Johannes Kepler [Fig. 2.1(b)] also witnessed a supernova explosion in A.D. 1604. These two events are now known as **Tycho’s supernova** and **Kepler’s supernova**, respectively.

Unfortunately, Kepler’s supernova was the last supernova observed to occur in the Milky Way Galaxy. However, on February 24.23 UT, 1987, Ian Shelton, using a 10-inch astrograph at the Las Campanas Observatory in Chile, detected **SN 1987A** just southwest of a massive molecular cloud region in the LMC known as 30 Doradus; the supernova is shown in Fig. 15.5. It was the first time since the development of modern instruments that a supernova had been seen so close to Earth (the distance to the LMC is 50 kpc). The excitement of the astronomical community worldwide was immediate and intense. It was quickly realized that the progenitor of this spectacular supernova was a blue supergiant star. Nicholas Sanduleak (1933–1990) had cataloged the star Sk –69 202 while investigating hot stars in the Magellanic Clouds.² The chance to observe a supernova from such a close vantage point using

²Sk –69 202 gets its name from being the 202nd entry in the –69° declination band of the Sanduleak catalog of stars in the Magellanic Clouds.

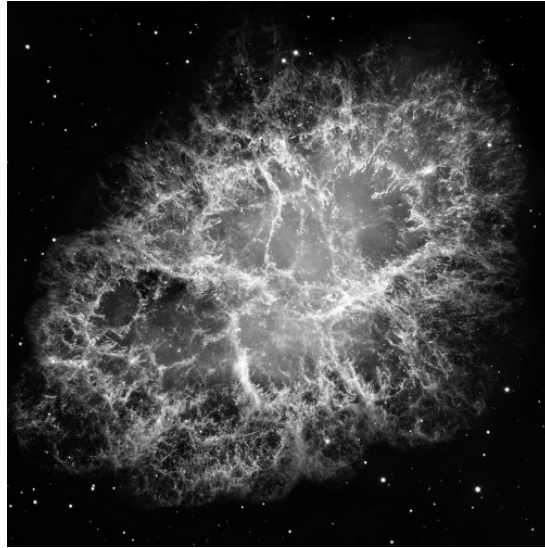


FIGURE 15.4 The Crab supernova remnant, located 2000 pc away in the constellation of Taurus. The remnant is the result of a Type II supernova that was observed for the first time on July 4, 1054. [Courtesy of NASA, ESA, J. Hester and A. Loll (Arizona State University).]



FIGURE 15.5 A portion of the Large Magellanic Cloud showing SN 1987A on the lower right-hand side of the photograph. 30 Doradus (also known as the Tarantula Nebula) is an immense H II region that is clearly evident on the left-hand side of the photograph. (Courtesy of the European Southern Observatory, ©ESO.)

the arsenal of tools available to modern astrophysics provided an ideal opportunity to test our theory of the fate of massive stars.

Classes of Supernovae

Today astronomers are able to routinely observe supernovae in other galaxies (hence the “A” in SN 1987A representing the first supernova reported that year). However, supernovae are exceedingly rare events, typically occurring about once every one hundred years or so in any one galaxy. As the spectra and light curves of supernovae have been carefully studied, it has been realized that there are several distinct classes of supernovae with different classes of underlying progenitors and mechanisms.

Type I supernovae were identified first as those supernovae that do not exhibit any hydrogen lines in their spectra. Given that hydrogen is the most abundant element in the universe, this fact alone suggests something unusual about these objects. Conversely, the spectra of **Type II** supernovae contain strong hydrogen lines.

Type I supernovae can be further subdivided according to their spectra. Those Type I spectra that show a strong Si II line at 615 nm are called **Type Ia**. The others are designated **Type Ib** or **Type Ic**, based on the presence (Ib) or absence (Ic) of strong helium lines.

Figure 15.6 shows examples of the spectra of each of the four types of supernovae (SNe) discussed here.

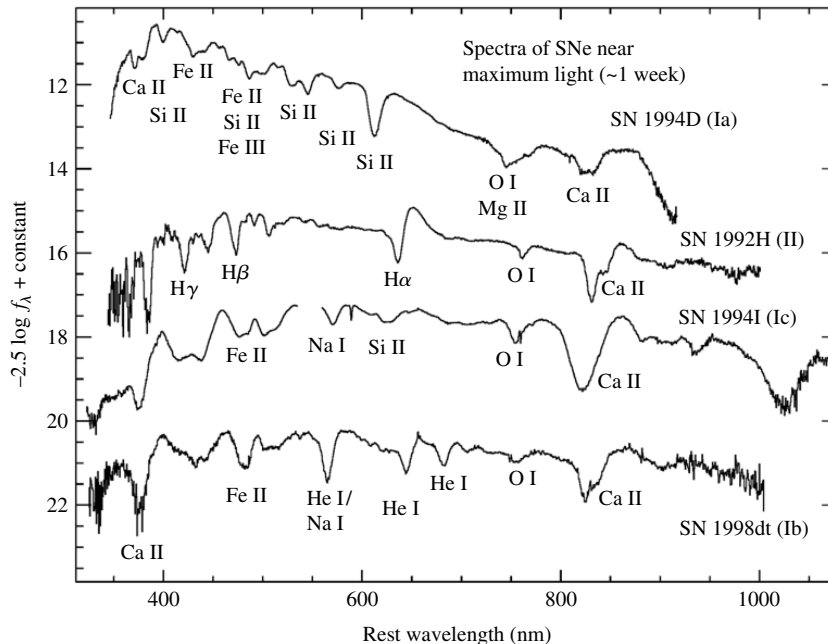


FIGURE 15.6 Representative spectra of the four types of supernovae; Type Ia, Ib, Ic, and II. Note that although SN 1994I (Type Ic) does exhibit a weak Si II absorption line, it is much less prominent than the Si lines in a Type Ia. Brightness is in arbitrary flux units. (Courtesy of Thomas Matheson, National Optical Astronomy Observatory.)

The lack of hydrogen lines in Type I supernovae indicates that the stars involved have been stripped of their hydrogen envelopes. The differences in the spectral signatures between Type Ia and Types Ib and Ic indicate that different physical mechanisms are at work. This is reflected in the different environments observed for these outbursts. Type Ia supernovae are found in all types of galaxies, including ellipticals that show very little evidence of recent star formation. On the other hand, Types Ib and Ic have been seen only in spiral galaxies, near sites of recent star formation (H II regions). This implies that short-lived massive stars are probably involved with Types Ib and Ic, but not with Type Ia.

Figure 15.7 shows a composite light curve at blue (B) wavelengths for Type I's. The typical peak brightness of a Type Ia is $M_B = -18.4$, while the light curves of Types Ib and Ic supernovae are fainter by 1.5 to 2 magnitudes in blue light but are otherwise similar. All Type I supernovae show similar rates of decline of their brightness after maximum, about $0.065 (\pm 0.007)$ magnitude per day at 20 days. After about 50 days, the rate of dimming slows and becomes constant, with Type Ia's declining 50% faster than the others (0.015 mag d^{-1} vs. 0.010 mag d^{-1}). It is believed that SN 1006 and the supernovae detected by Tycho (SN 1572) and Kepler (SN 1604) were Type I's.

Observationally, Type II supernovae are characterized by a rapid rise in luminosity, reaching a maximum brightness that is typically 1.5 mag dimmer than Type Ia's. The peak light output is followed by a steady decrease, dropping six to eight magnitudes in a year. Their spectra also exhibit lines associated with hydrogen and heavier elements. Furthermore, P Cygni profiles are common in many lines (indicating rapid expansion; see Section 12.3). The Crab supernova (SN 1054) and SN 1987A were Type II's.

The light curves of Type II supernovae can be classified as either **Type II-P** (plateau) or **Type II-L** (linear). Composite B magnitude light curves of each type are shown in Fig. 15.8. A temporary but clear plateau exists between about 30 and 80 days after maximum light for Type II-P supernovae; no such detectable plateau exists for Type II-L objects. Type II-P supernovae also occur approximately ten times as often as Type II-L's.

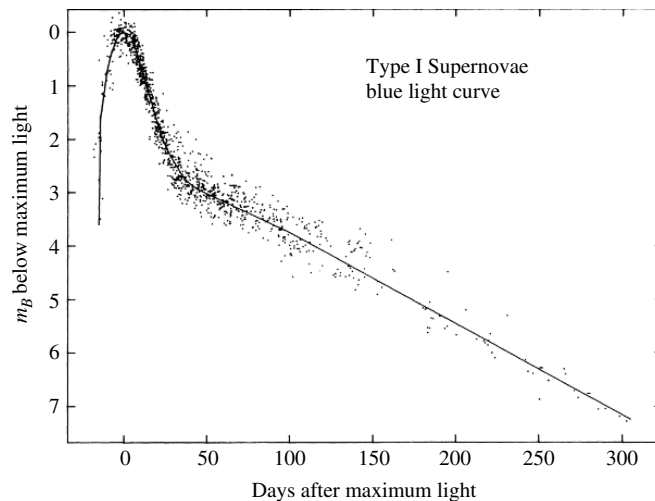


FIGURE 15.7 Composite light curve for Type I supernovae at blue wavelengths. All magnitudes are relative to m_B at maximum. (Figure adapted from Doggett and Branch, *Astron. J.*, 90, 2303, 1985.)

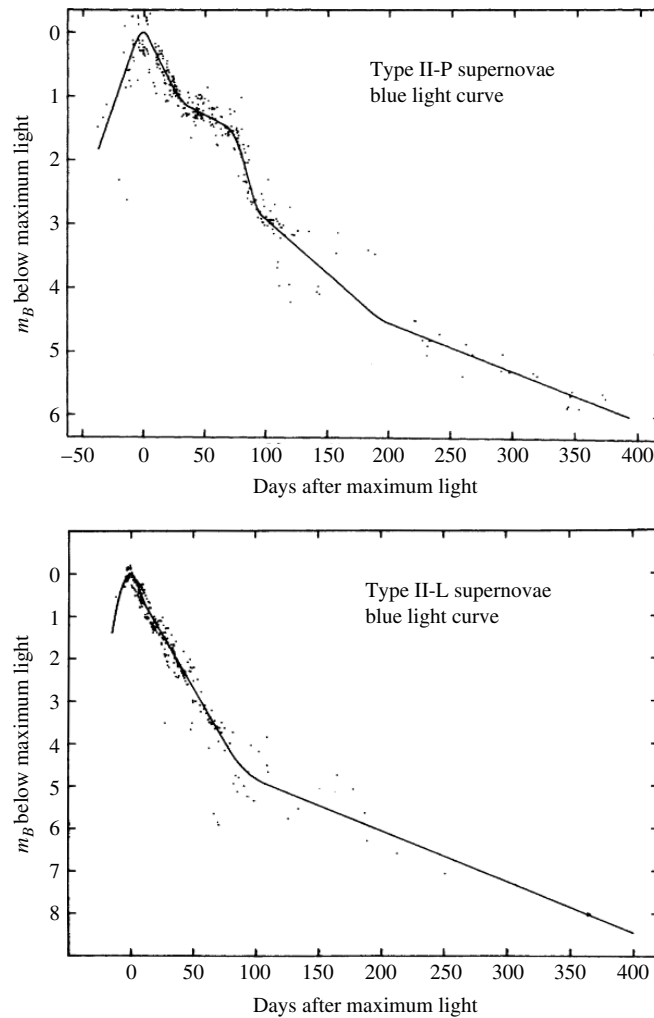


FIGURE 15.8 The characteristic shapes of Type II-P and Type II-L light curves. These are composite light curves, based on the observations of many supernovae. (Figures adapted from Doggett and Branch, *Astron. J.*, 90, 2303, 1985.)

A summary decision tree of supernova classification is given in Fig. 15.9.

Of course nature loves to confound our clean classification schemes. SN 1993J in the spiral galaxy M81 in Ursa Major initially displayed strong hydrogen emission lines (i.e., Type II), but within a month the hydrogen lines were replaced by helium and its appearance changed to that of a Type Ib. This provides some indication that at least Type Ib's and Type II's are related in some way. We will learn that Type Ic's are also closely related.

We know today that Type Ia's are fundamentally different events from other supernovae. In Section 15.3 we will discuss the physics involved in Types Ib, Ic, and II supernovae, postponing a detailed discussion of Type Ia's until Section 18.5.

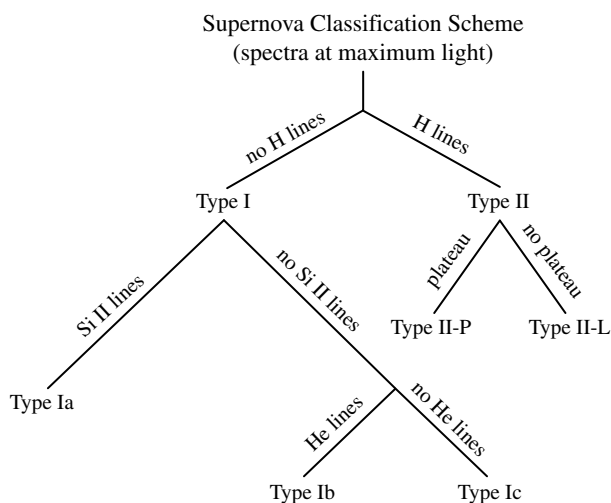


FIGURE 15.9 The classification of supernovae based on their spectra at maximum light and the existence or absence of a plateau in the Type II light curve.

15.3 ■ CORE-COLLAPSE SUPERNOVAE

The goal of understanding the physical processes involved in generating supernovae has been a long-standing challenge. The sheer amount of energy released in a supernova event is staggering. A typical Type II releases 10^{46} J of energy, with about 1% of that appearing as kinetic energy of the ejected material and less than 0.01% being released as the photons that produce the spectacular visual display. As we will see later, the remainder of the energy is radiated in the form of neutrinos. Similar values are obtained for Type Ib and Type Ic supernovae.

Example 15.3.1. To illustrate how much energy is involved in a Type II event, consider the equivalent amount of rest mass, and how much iron could ultimately be produced by releasing that much nuclear binding energy.

From $E = mc^2$, the energy released by a Type II supernova corresponds to a rest mass of

$$m = E/c^2 = 10^{46} \text{ J}/c^2 = 1 \times 10^{29} \text{ kg} = 0.06 M_{\odot}.$$

The binding energy of an iron-56 nucleus (${}^{56}_{26}\text{Fe}$), the most stable of all nuclei, is 492.26 MeV, and the mass of the nucleus is 55.934939 u (recall Fig. 10.9). In order to release 10^{46} J of energy through the formation of iron nuclei from protons and neutrons, it would be necessary to form

$$N = \left(\frac{1 \times 10^{46} \text{ J}}{492.26 \text{ MeV/nucleus}} \right) \left(\frac{1 \text{ MeV}}{1.6 \times 10^{-13} \text{ J}} \right) = 1.3 \times 10^{56} \text{ nuclei},$$

continued

corresponding to a mass of iron of

$$m = N(55.93 \text{ u})(1.66 \times 10^{-27} \text{ kg/u}) = 1.2 \times 10^{31} \text{ kg} = 5.9 M_{\odot}.$$

Of course this would require that $5.9 M_{\odot}$ of iron be produced all at once in an explosive event if this is the source of the energy for a Type II supernova! On the other hand, if this much iron were broken down to the original protons and neutrons, the equivalent amount of energy would need to be absorbed.

As we will see later, *iron is not formed* as a result of releasing the energy involved in a supernova explosion; in fact, the energy source is not nuclear. However, iron is critically involved in the process in a perhaps unexpected way.

Core-Collapse Supernova Mechanism

The post-main-sequence evolution of stars more massive than about $8 M_{\odot}$ is decidedly different from what was described in Chapter 13. Although hydrogen is converted into helium on the main sequence, followed by helium burning leading to a carbon–oxygen core, the very high temperature in the core of a massive star means that carbon and oxygen can burn as well. The end result is that rather than the star ending its life through the formation of a planetary nebula, a catastrophic supernova explosion occurs instead.

What follows is a discussion of that evolutionary process. Although all of the details have yet to be worked out at the time of writing, the story of how Type Ib, Type Ic, and Type II supernovae are produced is becoming clearer. The three types are all closely related and all involve the collapse of a massive, evolved stellar core. Hence, collectively Types Ib, Ic, and II are known as **core-collapse supernovae**.

As the helium-burning shell continues to add ash to the carbon–oxygen core, and as the core continues to contract, it eventually ignites in carbon burning, generating a variety of by-products, such as $^{16}_8\text{O}$, $^{20}_{10}\text{Ne}$, $^{23}_{11}\text{Na}$, $^{23}_{12}\text{Mg}$, and $^{24}_{12}\text{Mg}$ (see Eqs. 10.64–10.66). This leads to a succession of nuclear reaction sequences, the exact details of which depend sensitively on the mass of the star.

Assuming that each reaction sequence reaches equilibrium, an “onion-like” shell structure develops in the interior of the star. Following carbon burning, the oxygen in the resulting neon–oxygen core will ignite (Eq. 10.67), producing a new core composition dominated by $^{28}_{14}\text{Si}$. Finally, at temperatures near $3 \times 10^9 \text{ K}$, silicon burning can commence through a series of reactions such as



$$\vdots$$


Silicon burning produces a host of nuclei centered near the $^{56}_{26}\text{Fe}$ peak of the binding energy per nucleon curve shown in Fig. 10.9, the most abundant of which are probably $^{54}_{26}\text{Fe}$, $^{56}_{26}\text{Fe}$, and $^{56}_{28}\text{Ni}$. Any further reactions that produce nuclei more massive than $^{56}_{26}\text{Fe}$ are endothermic